

The Anatomy of Machine Learning-Based Portfolio Performance

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Abstract

Asset return predictability is routinely assessed by economic value: based on a set of predictors, out-of-sample return forecasts are generated—increasingly via “black box” machine learning models—which serve as inputs for portfolio construction, and performance metrics are computed over an evaluation period. We develop a flexible methodology based on Shapley values—the *Shapley-based portfolio performance contribution* (SPPC)—to exactly allocate the contributions of individual or groups of predictors to a performance metric. We illustrate the SPPC in an empirical application measuring the economic value of cross-sectional stock return predictability using a large number of firm characteristics and machine learning.

JEL classifications: C53, C55, C58, G11, G17

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1. Introduction

Asset return predictability is a leading topic in empirical asset pricing. Out-of-sample tests are now routinely employed, as they are viewed as the most rigorous and informative tests of return predictability, particularly in the era of big data and machine learning (e.g., Nagel 2021; Martin and Nagel 2022). In terms of analyzing out-of-sample return predictability, in addition to assessing the statistical accuracy of return forecasts,¹ it is now routine to analyze the *economic value* of return predictability via asset allocation exercises. Specifically, return forecasts based on a set of predictors serve as inputs for constructing a portfolio. Portfolio performance metrics are then computed over a forecast evaluation period (and often compared to those for a benchmark portfolio) to measure the economic value of return predictability from an investment perspective.

¹For example, in the context of aggregate equity market return predictability, the popular out-of-sample R^2 statistic (Fama and French 1989; Campbell and Thompson 2008) measures the proportional reduction in mean squared error for a competing forecast based on the information in a set of predictors vis-à-vis a naïve benchmark forecast that ignores the information. Han et al. (2024) develop a modified out-of-sample R^2 statistic for analyzing cross-sectional return predictability.

A spate of recent studies employs a multitude of firm characteristics and machine learning methods to generate cross-sectional out-of-sample stock return forecasts (e.g., Freyberger, Neuhierl, and Weber 2020; Gu, Kelly, and Xiu 2020; Avramov, Cheng, and Metzker 2023; Han et al. 2024). They construct long-short portfolios by sorting stocks according to their return forecasts for the next month and going long (short) stocks with the highest (lowest) return forecasts. Similarly to studies of aggregate market return predictability,² these studies find that long-short portfolios based on machine learning forecasts provide substantive economic value to investors as reflected by portfolio performance metrics (especially Sharpe ratios), thereby furnishing strong evidence of cross-sectional stock return predictability.

However, the existing literature does not provide a general methodology for measuring the contributions of predictors in machine learning models to portfolio performance that produces an exact decomposition of a performance metric. In the present paper, we fill this gap by developing a methodology based on Shapley (1953) values to allocate the contributions of individual or groups of predictors to a portfolio performance metric. This allows us to exactly decompose portfolio performance in terms of the underlying predictors—in essence, we “anatomize” machine learning-based portfolio performance. Based on the logic of Shapley values, our methodology provides a framework for fairly allocating the contributions of predictors in fitted prediction models with respect to the economic value that they generate. We call our new measure the *Shapley-based portfolio performance contribution*, which we denote for predictor p by SPPC_p .

Our portfolio performance decomposition based on the SPPC_p can be viewed as a machine learning model interpretation tool for finance. With the growing popularity of machine learning models, many of which are “black boxes,” numerous model interpretation tools have been developed in the machine learning literature, including attribution and variable importance metrics as well as measures of interactions and nonlinearities (e.g., Dimopoulos, Bourret, and Lek 1995; Friedman 2001; Štrumbelj and Kononenko 2010, 2014; Goldstein et al. 2015;

²See Rapach and Zhou (2022) for a review of the literature on aggregate equity market return predictability, including its economic value.

Ribeiro, Singh, and Guestrin 2016; Lundberg and Lee 2017; Greenwell, Boehmke, and McCarthy 2018; Fisher, Rudin, and Dominici 2019; Apley and Zhu 2020; Covert, Lundberg, and Lee 2020).³ Existing model interpretation tools are primarily designed to analyze fitted prediction models based on training sample data. While conventional model interpretation tools are informative for investigating the relevance of predictors in fitted models, in the context of analyzing the economic value of return predictability, they are not designed to directly measure how predictors influence portfolio performance per se, which is the ultimate object of interest in our context. Our new SPPC_p does this for any model in a manner that produces an exact decomposition of portfolio performance, thereby providing a valuable model interpretation tool for deepening our understanding of the roles of individual or groups of predictors in fitted machine learning models when it comes to the economic value of return predictability.⁴

In developing the SPPC_p , we begin by reviewing the use of Shapley values for interpreting fitted prediction models (Štrumbelj and Kononenko 2010, 2014; Lundberg and Lee 2017). We use Shapley values as a foundation for our new metric, as they possess a number of properties (e.g., local accuracy, missingness, symmetry, and linearity) that provide an especially attractive model interpretation framework. To fix ideas, we focus on a setting where we forecast individual stock returns based on firm characteristics using a pooled prediction model, as in Freyberger, Neuhierl, and Weber (2020), Gu, Kelly, and Xiu (2020), and Avramov, Cheng, and Metzker (2023). While we focus on this setting, the SPPC_p can be computed for any situation in which we use a fitted prediction model (or ensemble of models) to forecast asset returns, with the return forecasts serving as inputs for constructing a portfolio.

³See Molnar (2025) for a textbook treatment of machine learning model interpretation tools.

⁴Moehle, Boyd, and Ang (2022) propose tools, some of which are based on Shapley values, that are designed to attribute portfolio performance to “features” such as rebalancing frequency, leverage limits, and ESG constraints. The SPPC_p is very different, as it measures the contributions of the predictors in fitted machine learning models to portfolio performance, thereby providing insight into the underlying sources of the economic value generated by return predictability. Aleti, Bollerslev, and Sigggaard (2026) use a Shapley-based procedure to measure the contributions of predictors to portfolio performance in their study of intraday return predictability. Our methodology differs significantly for their procedure (e.g., their procedure is limited to linear prediction models, while our approach is model agnostic, so it accommodates nonlinear machine learning models).

We explain how we extend conventional Shapley values to measure the contributions of predictors to (1) an out-of-sample return forecast, (2) a portfolio return, and (3) a portfolio performance metric, resulting in the SPPC_p . The SPPC_p inherits all of the attractive properties of Shapley values. We emphasize that the SPPC_p is very flexible: it is model agnostic (i.e., it applies to any prediction model, including all types of machine learning models), it can be used for any strategy for mapping return forecasts to portfolio weights, and it can be computed for any performance metric.⁵ We use an efficient sampling algorithm to compute the SPPC_p .

We illustrate the use of the SPPC_p in an extensive empirical application investigating the economic value of forecasting individual stock returns using a machine learning model and 207 firm characteristics from Chen and Zimmermann (2022). We generate monthly out-of-sample firm-level stock return forecasts via the **XGBoost** algorithm (Chen and Guestrin 2016), a powerful machine learning technique based on gradient boosting decision trees that performs well in forecasting competitions in a variety of domains. We use the XGBoost return forecasts to sort stocks into quintiles and construct a zero-investment portfolio that goes long (short) the fifth (first) quintile, with each leg value-weighted. The long-short portfolio based on the XGBoost forecasts performs impressively, generating annualized Sharpe and Calmar ratios of 1.80 and 1.44, respectively, for the 1973:01 to 2021:12 forecast evaluation period, both of which are well above the corresponding ratios for the aggregate market portfolio (0.47 and 0.14, respectively). The XGBoost portfolio also generates economically large and statistically significant alphas in the context of two leading multifactor models, namely, the Fama and French (2018) six-factor and Hou et al. (2021) augmented q-factor models. In sum, we find that firm-level stock return forecasts based on a large number of firm characteristics and machine learning produce substantial economic value.

⁵The SPPC_p can also be used to measure the contributions of predictors to portfolio performance when machine learning approaches are used to directly estimate optimal portfolio weights (e.g., Kozak, Nagel, and Santosh 2020; Cong et al. 2022; Chen, Pelger, and Zhu 2024; Jensen et al. 2026).

After placing the individual firm characteristics into 20 groups based on economic concepts, we estimate the contributions of the predictor groups to portfolio performance using the SPPC_p . The **Risk**, **Earnings**, **Seasonal momentum**, and **Momentum** groups play leading roles in accounting for the substantive Sharpe and Calmar ratios as well as the sizable alphas generated by the XGBoost portfolio. For example, the XGBoost portfolio increases the Sharpe ratio by 1.33 vis-à-vis the market portfolio; the four groups together account for 0.89 (67%) of the increase. In contrast, the **Sales**, **Ownership**, and **Reversal** groups often substantively detract from portfolio performance across the different metrics.

The performance of the XGBoost portfolio tends to diminish after the early 2000s, although it still outperforms the market portfolio, especially during business-cycle recessions. To examine how the contributions of the predictor groups to portfolio performance change over time, we use the SPPC_p to estimate the group contributions for 60-month rolling windows from the full forecast evaluation period. While the **Risk** and **Momentum** groups typically make substantial positive contributions to portfolio performance through the early 2000s, they often make negative contributions thereafter. Groups making consistently positive and sizable contributions over the full forecast evaluation period include **Earnings**, **Seasonal momentum**, and **Investment**. Overall, the SPPC_p sheds light on how the predictor groups contribute to portfolio performance, thereby improving our understanding of the role of predictors in generating economic value with respect to cross-sectional return predictability in a machine learning model. As such, the SPPC_p is a valuable tool for identifying the key determinants of cross-sectional expected returns in terms of economic value.

Finally, we compare the SPPC_p to an alternative approach that measures the contribution of a predictor group by breaking the connection between the predictor group and the fitted model predictions. This approach is intuitive and less computationally costly than our Shapley-based SPPC_p . However, it is not based on the logic of Shapley values and does not share their properties; for example, unlike our SPPC_p , it does not provide an exact decomposition of a performance metric. Furthermore, for our application, we find that

the estimated contributions of the predictor groups to portfolio performance often differ substantially across the two approaches. Overall, because the two approaches generally do not provide similar results—and the SPPC_p follows the logic and has the attractive properties of Shapley values—we view the benefits accruing to the SPPC_p as outweighing the increase in computational cost.

The rest of the paper is organized as follows. Section 2 provides background on the conventional use of Shapley values for model interpretation in the machine learning literature and explains our extensions, culminating in the SPPC_p . Section 3 reports results for the empirical application. Section 4 concludes.

2. Methodology

This section presents our methodology for decomposing portfolio performance in terms of the underlying predictors that guide asset allocation using the SPPC_p . In line with our application in Section 3, we consider a setting in which we forecast individual stock returns using firm characteristics via a pooled prediction model. To simplify the presentation, we focus on regression (i.e., predicting a real-valued number). The methodology and notation can be straightforwardly tailored for classification (i.e., predicting a class or category), and all the results in this section hold for both regression and classification.

2.1. Shapley Values

As background, we begin with a description of conventional Shapley values, which are viewed as one of, if not the most, informative model interpretation tools in machine learning. Shapley (1953) values are a canonical solution method for fairly allocating the total payoff to the players in a coalitional game. The intuition for using Shapley values for model interpretation is to exploit the analogy between players in a cooperative game earning a payoff and predictors in a model making a prediction. According to the logic of Shapley values, payoffs are fairly allocated to the players in a game. In the context of prediction, we are interested

in fairly allocating the contributions of the predictors to a fitted model’s prediction. This is a nontrivial task, especially for models with interactions between predictors and nonlinearities. Štrumbelj and Kononenko (2010, 2014) and Lundberg and Lee (2017) show how Shapley values can be used to allocate the contributions of the predictors to a prediction made by the model. We adapt their ideas to a panel setting where a model generates predictions for individual stock returns over time based on a set of firm characteristics. While we focus on this setting to fix ideas and because of its use in recent studies, we emphasize that our methodology can be applied to any prediction model.

In terms of notation, we index individual predictors by p , with the index set of predictors denoted by $S = \{1, \dots, P\}$. We index cross-sectional units by i and denote the index set of cross-sectional units by $C = \{1, \dots, N\}$.⁶ We denote the P -vector of firm characteristics (i.e., predictors) for stock i in period t by $\mathbf{x}_{i,t} = [x_{1,i,t} \ \dots \ x_{P,i,t}]'$, while $r_{i,t}$ denotes the return on stock i in period t . The prediction model is given by

$$r_{i,t+1} = \underbrace{f(\mathbf{x}_{i,t})}_{\substack{\text{conditional} \\ \text{expectation} \\ \text{function}}} + \varepsilon_{i,t+1}, \quad (1)$$

where $\varepsilon_{i,t}$ is a zero-mean disturbance term. We use $W_j = \{t_{j,\text{start}}, \dots, t_{j,\text{end}} - 1\}$ to represent the window of panel data observations used to train the model. The prediction model can be estimated using an expanding or rolling window along the time dimension; for the former (latter), the cardinality of W_j increases (remains constant). We denote the fitted prediction function evaluated at instance $\mathbf{x}_{i,t}$ and trained using window W_j by $\hat{f}(\mathbf{x}_{i,t}; W_j)$.

The Shapley value measures the marginal contribution of the predictor $x_{p,i,t}$ to the prediction $\hat{f}(\mathbf{x}_{i,t}; W_j)$ given $S \setminus \{p\}$ (i.e., given the presence of all of the other predictors in the model). By relying on insights from coalitional game theory, Shapley values fairly allocate the marginal contributions among the individual predictors. Formally, we can express the

⁶For notational simplicity, we assume that the number of cross-sectional units is the same each period. In our empirical application in Section 3, the number of cross-sectional units (i.e., firms) changes over time. It is straightforward to modify the notation to allow for time variation in the number of cross-sectional units.

Shapley value for predictor p and instance $\mathbf{x}_{i,t}$ for a prediction model trained using the panel data observations in window W_j as

$$\phi_p(\mathbf{x}_{i,t}; W_j) = \sum_{Q \subseteq S \setminus \{p\}} \underbrace{\frac{|Q|!(P - |Q| - 1)!}{P!}}_{\text{coalition } Q\text{'s weight}} \underbrace{[\xi_{Q \cup \{p\}}(\mathbf{x}_{i,t}; W_j) - \xi_Q(\mathbf{x}_{i,t}; W_j)]}_{\text{predictor } p\text{'s marginal contribution}} \quad (2)$$

for $p \in S$, $i \in C$, and $t \in W_j$, where Q is a subset of predictors (i.e., a coalition), $Q \subseteq S \setminus \{p\}$ constitutes the set of all possible coalitions of $P - 1$ predictors in S that exclude predictor p , $|Q|$ is the cardinality of Q , and $\xi_Q(\mathbf{x}_{i,t}; W_j)$ is the value function. The expression in brackets on the right-hand-side of Equation (2) is the change in the value function for the predictors in the coalition Q in conjunction with predictor p relative to the predictors in Q only, which measures predictor p 's marginal contribution corresponding to the coalition Q . Equation (2) takes a weighted average of the changes in the value function for all possible coalitions of $P - 1$ predictors that exclude p . The change in the value function receives the weight $|Q|!(P - |Q| - 1)!/P!$, where the weights sum to one. In sum, to measure the contribution of p to the prediction corresponding to observation $\mathbf{x}_{i,t}$, Shapley values rely on coalitions to control for the presence of the other predictors in the model.

In the machine learning literature, different value functions—in terms of how predictors not in a coalition are removed—define different Shapley values (e.g., Chen et al. 2020, 2023). We use the *interventional* Shapley value (Datta, Sen, and Zick 2016; Janzing, Minorics, and Blöbaum 2020; Sundararajan and Najmi 2020), for which the value function is defined as

$$\xi_Q(\mathbf{x}_{i,t}; W_j) = \mathbb{E}_{\mathbb{P}(x_{l,i,t} : l \notin Q)} [\hat{f}(x_{k,i,t} : k \in Q, x_{l,i,t} : l \notin Q; W_j)]. \quad (3)$$

Equation (3) removes the predictors not in the coalition Q via their marginal distribution. In the taxonomy of Chen et al. (2020), the interventional Shapley value is “true to the model.” This aligns with our goal, as we are interested in interpreting the specific fitted model that generates the return predictions.⁷

⁷Alternatively, for the *observational* Shapley value, the value function in Equation (2) removes the predictors not in the coalition Q via their conditional distribution (i.e., conditional on the predictors in Q). Chen et al. (2020) provide an insightful philosophical discussion of observational versus interventional Shapley values that supports the use of the latter in our context.

The interventional Shapley value in Equations (2) and (3) is characterized by a number of attractive properties, including the following:

- **Efficiency**

$$\sum_{p \in S} \phi_p(\mathbf{x}_{i,t}; W_j) = \hat{f}(\mathbf{x}_{i,t}; W_j) - \underbrace{\mathbb{E}[\hat{f}; W_j]}_{\text{baseline prediction}} \quad (4)$$

for $i \in C$ and $t \in W_j$, where the baseline prediction corresponds to the unconditional expectation of $\hat{f}$.⁸

- **Missingness**

$$\forall R \subseteq S \setminus \{p\} : \xi_{R \cup \{p\}}(\mathbf{x}_{i,t}; W_j) = \xi_R(\mathbf{x}_{i,t}; W_j) \Rightarrow \phi_p(\mathbf{x}_{i,t}; W_j) = 0. \quad (5)$$

- **Symmetry**

$$\begin{aligned} \forall R \subseteq S \setminus \{p, q\} : \xi_{R \cup \{p\}}(\mathbf{x}_{i,t}; W_j) = \xi_{R \cup \{q\}}(\mathbf{x}_{i,t}; W_j) \Rightarrow \\ \phi_p(\mathbf{x}_{i,t}; W_j) = \phi_q(\mathbf{x}_{i,t}; W_j). \end{aligned} \quad (6)$$

- **Linearity**

For any real numbers c_1 and c_2 and models $\hat{f}(\mathbf{x}_{i,t}; W_j)$ and $\hat{f}'(\mathbf{x}_{i,t}; W_j)$,

$$\phi_p(c_1[\hat{f}(\mathbf{x}_{i,t}; W_j) + c_2\hat{f}'(\mathbf{x}_{i,t}; W_j)]) = c_1\phi_p(\hat{f}(\mathbf{x}_{i,t}; W_j)) + c_1c_2\phi_p(\hat{f}'(\mathbf{x}_{i,t}; W_j)). \quad (7)$$

According to efficiency (also known as *local accuracy*) in Equation (4), the model prediction corresponding to observation $\mathbf{x}_{i,t}$ can be exactly decomposed (in terms of the deviation from the baseline prediction) into the sum of the Shapley values for the individual predictors for that observation. Missingness ensures that predictors that are not used by the fitted model are not attributed any importance.⁹ Symmetry and linearity are intuitively appealing

⁸This is a natural baseline for a prediction model, as it is the forecast based on the empty coalition set.

⁹Missingness generally does not hold for the observational Shapley value in Footnote 7 (Janzing, Minorics, and Blöbaum 2020; Sundararajan and Najmi 2020), which further motivates our choice of the interventional Shapley value.

properties, and the latter is useful for computing Shapley values for ensembles of prediction models.

It becomes practically infeasible to exactly compute the Shapley value in Equation (2) as the number of predictors becomes large. The issue is that the number of coalitions Q becomes very large as P increases. Štrumbelj and Kononenko (2010, 2014) propose an algorithm based on the sampling approximation of Castro, Gómez, and Tejada (2009). We develop a refined version of their algorithm to estimate the Shapley value in Equation (2). In Section 2.2, we extend the algorithm to estimate the contributions of the individual predictors to a portfolio performance metric.

We begin by expressing Equation (2) in the following equivalent form:

$$\phi_p(\mathbf{x}_{i,t}; W_j) = \frac{1}{P!} \sum_{\mathcal{O} \in \pi(P)} [\xi_{\text{Pre}_p(\mathcal{O}) \cup \{p\}}(\mathbf{x}_{i,t}; W_j) - \xi_{\text{Pre}_p(\mathcal{O})}(\mathbf{x}_{i,t}; W_j)] \quad (8)$$

for $p \in S$, $i \in C$, and $t \in W_j$, where $\mathcal{O}$ is an ordered permutation for the predictor indices in S , $\pi(P)$ is the set of all ordered permutations for S , and $\text{Pre}_p(\mathcal{O})$ is the set of indices that precede p in $\mathcal{O}$. To implement the algorithm, we make a random draw m with replacement of an ordered permutation from $\pi(P)$, denoted by $\mathcal{O}_m$. Using $\mathcal{O}_m$, we compute

$$\begin{aligned} \hat{\theta}_{p,m}(\mathbf{x}_{i,t}; W_j) = & \underbrace{\frac{1}{|C||W_j|} \sum_{u \in C} \sum_{s \in W_j} \hat{f}(x_{k,i,t} : k \in \text{Pre}_p(\mathcal{O}_m) \cup \{p\}, x_{l,u,s} : l \in \text{Post}_p(\mathcal{O}_m); W_j)}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m) \cup \{p\}}(\mathbf{x}_{i,t}; W_j)} - \\ & \underbrace{\frac{1}{|C||W_j|} \sum_{u \in C} \sum_{s \in W_j} \hat{f}(x_{k,i,t} : k \in \text{Pre}_p(\mathcal{O}_m), x_{l,u,s} : l \in \text{Post}_p(\mathcal{O}_m) \cup \{p\}; W_j)}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m)}(\mathbf{x}_{i,t}; W_j)} \end{aligned} \quad (9)$$

for $p \in S$, $i \in C$, and $t \in W_j$, where $\text{Post}_p(\mathcal{O})$ is the set of indices that follow p in $\mathcal{O}$. Observe that the value function estimates on the right-hand-side of Equation (9) remain true to the model by integrating out the predictors not in a coalition via Monte Carlo integration using “background data” from the training sample W_j , consistent with the value function in Equation (3) for the interventional Shapley value.

Using Equation (9), the sampling approximation of the Shapley value in Equation (8) is given by

$$\hat{\phi}_p(\mathbf{x}_{i,t}; W_j) = \frac{1}{2M} \sum_{m=1}^{2M} \hat{\theta}_{p,m}(\mathbf{x}_{i,t}; W_j) \quad (10)$$

for $p \in S$, $i \in C$, and $t \in W_j$, where M is the number of draws. We employ antithetic sampling as a variance-reduction device in Equation (10) by computing $\hat{\theta}_{p,m}(\mathbf{x}_{i,t}; W_j)$ in Equation (9) for the original order of a randomly drawn ordered permutation and when the order is reversed (Mitchell et al. 2022). The sampling approximation $\hat{\phi}_p(\mathbf{x}_{i,t}; W_j)$ in Equation (10) retains all of the properties of $\phi_p(\mathbf{x}_{i,t}; W_j)$ in Equation (8) listed in Equations (4) to (7), including local accuracy:

$$\sum_{p \in S} \hat{\phi}_p(\mathbf{x}_{i,t}; W_j) = \hat{f}(\mathbf{x}_{i,t}; W_j) - \underbrace{\bar{\hat{f}}(W_j)}_{\hat{\phi}_\emptyset(W_j)} \quad (11)$$

for $i \in C$ and $t \in W_j$, where

$$\bar{\hat{f}}(W_j) = \frac{1}{|C||W_j|} \sum_{i \in C} \sum_{t \in W_j} \hat{f}(\mathbf{x}_{i,t}; W_j) \quad (12)$$

is the average in-sample prediction for the model trained using sample W_j . The average in-sample prediction corresponds to the baseline prediction based on the empty coalition set, denoted by $\hat{\phi}_\emptyset(W_j)$ in Equation (11).¹⁰

2.2. Decomposing Portfolio Performance

To this point, we have followed the convention of computing Shapley values for the fitted model predictions for the training sample observations. For developing the SPPC_p , we begin by defining the Shapley value corresponding to an *out-of-sample prediction*. Suppose that we train a model using window W_j and generate an out-of-sample return forecast for stock

¹⁰To ensure that local accuracy holds for $\hat{\phi}_p(\mathbf{x}_{i,t}; W_j)$ in Equation (10), we need to compute $\hat{\theta}_{p,m}(\mathbf{x}_{i,t}; W_j)$ in Equation (9) for each predictor $p \in S$ for a given random draw m (Castro, Gómez, and Tejada 2009, Proposition 3.2).

i and period $t_{j,\text{end}} + 1$ based on the fitted model:

$$\hat{r}_{i,t_{j,\text{end}}+1} = \hat{f}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) \quad (13)$$

for $i \in C$. We define the Shapley value corresponding to $\hat{r}_{i,t_{j,\text{end}}+1}$ in Equation (13) as

$$\phi_p^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) = \frac{1}{P!} \sum_{\mathcal{O} \in \pi(P)} [\xi_{\text{Pre}_p(\mathcal{O}) \cup \{p\}}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) - \xi_{\text{Pre}_p(\mathcal{O})}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)] \quad (14)$$

for $p \in S$ and $i \in C$. We suitably modify the algorithm to compute the sampling approximation to $\phi_p^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)$ in Equation (14). We make a random draw m and compute

$$\begin{aligned} \hat{\theta}_{p,m}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) = & \underbrace{\frac{1}{|C||W_j|} \sum_{u \in C} \sum_{s \in W_j} \hat{f}(x_{k,i,t_{j,\text{end}}} : k \in \text{Pre}_p(\mathcal{O}_m) \cup \{p\}, x_{l,u,s} : l \in \text{Post}_p(\mathcal{O}_m) ; W_j)}_{\hat{r}_{i,t_{j,\text{end}}+1,m,p}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) := \hat{\xi}_{\text{Pre}_p(\mathcal{O}_m) \cup \{p\}}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)} \\ & - \underbrace{\frac{1}{|C||W_j|} \sum_{u \in C} \sum_{s \in W_j} \hat{f}(x_{k,i,t_{j,\text{end}}} : k \in \text{Pre}_p(\mathcal{O}_m), x_{l,u,s} : l \in \text{Post}_p(\mathcal{O}_m) \cup \{p\} ; W_j)}_{\hat{r}_{i,t_{j,\text{end}}+1,m,\setminus p}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) := \hat{\xi}_{\text{Pre}_p(\mathcal{O}_m)}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)} \end{aligned} \quad (15)$$

for $p \in S$ and $i \in C$; the sampling approximation to $\phi_p^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)$ is given by

$$\hat{\phi}_p^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) = \frac{1}{2M} \sum_{m=1}^{2M} \hat{\theta}_{p,m}^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) \quad (16)$$

for $p \in S$ and $i \in C$. Note that value function estimates on the right-hand-side of Equation (15) continue to integrate out the predictors not in a coalition using background data from the training sample W_j , so we remain true to the model that produces the out-of-sample prediction.

The sampling approximation to the Shapley value for the out-of-sample prediction in Equation (16) continues to be characterized by local accuracy (as well as the other properties of Shapley values in Section 2.1):

$$\sum_{p \in S} \hat{\phi}_p^{\text{OS}}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j) = \underbrace{\hat{f}(\mathbf{x}_{i,t_{j,\text{end}}} ; W_j)}_{\hat{r}_{i,t_{j,\text{end}}+1}} - \hat{\phi}_{\emptyset}(W_j) \quad (17)$$

for $i \in C$. Equation (17) says that we can exactly decompose the one-period-ahead out-of-sample return forecast for stock i (in terms of the deviation from the empty coalition or baseline forecast) into the contributions of the P predictors.

Next, consider an investor who decides on their allocations across the N stocks for period $t_{j,\text{end}} + 1$ based on the set of return forecasts formed using data through period $t_{j,\text{end}}$. The allocation to stock i generally depends on the entire set of return forecasts for $t_{j,\text{end}} + 1$, so we denote the portfolio weight for i by the function

$$w_{i,t_{j,\text{end}}+1}(\{\hat{f}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C}) \quad (18)$$

for $i \in C$. The investor could, for example, employ a portfolio optimizer based on the return forecasts or form a long-short portfolio by going long (short) the stocks with the highest (lowest) return forecasts. Our methodology is general, so it applies to any strategy for mapping the return forecasts to the portfolio weights.

The portfolio return for $t_{j,\text{end}} + 1$ is given by

$$r_{t_{j,\text{end}}+1}^{\text{Port}} = \sum_{i \in C} w_{i,t_{j,\text{end}}+1}(\{\hat{f}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C}) r_{i,t_{j,\text{end}}+1}. \quad (19)$$

We define the Shapley value corresponding to the *out-of-sample portfolio return* for period $t_{j,\text{end}} + 1$:

$$\begin{aligned} \phi_p^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j) &= \frac{1}{P!} \sum_{\mathcal{O} \in \pi(P)} [\xi_{\text{Pre}_p(\mathcal{O}) \cup \{p\}}^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j) - \\ &\quad \xi_{\text{Pre}_p(\mathcal{O})}^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j)] \end{aligned} \quad (20)$$

for $p \in S$. Using the sampling algorithm, we make a random draw m and compute

$$\begin{aligned} \hat{\theta}_{p,m}^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j) &= \underbrace{\sum_{i \in C} [w_{i,t_{j,\text{end}}+1}(\{\hat{r}_{i,t_{j,\text{end}}+1,m,p}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C}) r_{i,t_{j,\text{end}}+1}]}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m) \cup \{p\}}^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j)} - \\ &\quad \underbrace{\sum_{i \in C} [w_{i,t_{j,\text{end}}+1}(\{\hat{r}_{i,t_{j,\text{end}}+1,m,\setminus p}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C}) r_{i,t_{j,\text{end}}+1}]}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m)}^{\text{PR}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W_j)} \end{aligned} \quad (21)$$

for $p \in S$, where $\hat{r}_{i,t_j,\text{end}+1,m,p}(\mathbf{x}_{i,t_j,\text{end}}; W_j)$ and $\hat{r}_{i,t_j,\text{end}+1,m,\setminus p}(\mathbf{x}_{i,t_j,\text{end}}; W_j)$ are defined in Equation (15); because Equation (15) relies on Monte Carlo integration using background data from the training sample W_j , we continue to remain true to the model. Using Equation (21), the sampling approximation to $\phi_p^{\text{PR}}(\{\mathbf{x}_{i,t_j,\text{end}}\}_{i \in C}; W_j)$ in Equation (20) is given by

$$\hat{\phi}_p^{\text{PR}}(\{\mathbf{x}_{i,t_j,\text{end}}\}_{i \in C}; W_j) = \frac{1}{2M} \sum_{m=1}^{2M} \hat{\theta}_{p,m}^{\text{PR}}(\{\mathbf{x}_{i,t_j,\text{end}}\}_{i \in C}; W_j) \quad (22)$$

for $p \in S$.

An important issue in implementing the Shapley-based decomposition of the portfolio return is deciding on the return corresponding to the empty coalition set. With regard to the empty coalition set, we have no predictors to determine the portfolio weights in Equation (18), so we need to specify the portfolio return for the empty coalition set, which we denote by $r_{t_j,\text{end}+1}^{\text{Base}}$, since the portfolio can be viewed as a “baseline” portfolio. We need $r_{t_j,\text{end}+1}^{\text{Base}}$ to compute Equation (21) when p is the first element in the ordered permutation $\mathcal{O}_m$; $r_{t_j,\text{end}+1}^{\text{Base}}$ also appears in the local accuracy condition in Equation (23) below.

The selection of the baseline portfolio is at the discretion of the researcher. To determine the baseline portfolio, it is sensible to ask, “If I had an empty set of predictors—and so no predictor information—how would I form a portfolio?” A relevant baseline will depend on the context; for example, for a portfolio that broadly invests in equities, the CRSP value-weighted aggregate market portfolio seems a natural choice. In addition to the other properties of Shapley values, Equation (22) satisfies local accuracy:

$$\sum_{p \in S} \hat{\phi}_p^{\text{PR}}(\{\mathbf{x}_{i,t_j,\text{end}}\}_{i \in C}; W_j) = r_{t_j,\text{end}+1}^{\text{Port}} - r_{t_j,\text{end}+1}^{\text{Base}}. \quad (23)$$

Equation (23) indicates that we can exactly decompose the portfolio return in period $t_j,\text{end}+1$ (in terms of the deviation from the baseline portfolio return) into the return contributions made by each of the predictors.

Finally, we extend our approach to compute the SPPC_p . To do so, we need to take into account the entire series of out-of-sample return forecasts and corresponding portfolio

returns over the forecast evaluation period. In terms of the time dimension, we assume that the sample of panel data spans T periods and that data through period T_{in} are used to train the model that generates the first set of out-of-sample return forecasts for period $T_{\text{in}} + 1$. The model is then retrained using panel data through $T_{\text{in}} + 1$ to generate the next set of return forecasts for $T_{\text{in}} + 2$. Continuing in this manner through the end of the available sample, $T - T_{\text{in}} = D$ sets of return forecast are generated, where the final model is trained using panel data through $T - 1$ to compute the last set of return forecasts for T . We define the index set of training windows used to fit the sequence of models as $W = \{1, \dots, D\}$, where $t_{j,\text{end}}$ corresponds to $T_{\text{in}}, T_{\text{in}} + 1, \dots, T - 1$ for $j = 1, 2, \dots, D$, respectively.

The key insight for computing the SPPC_p is to wrap a function corresponding to the performance metric around the portfolio return series. Denoting a performance metric function by $\mathcal{M}(\cdot)$, which depends on the sequence of portfolio returns over the forecast evaluation period, the Shapley value corresponding to the *portfolio performance metric* is defined by

$$\phi_p^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M}) = \frac{1}{P!} \sum_{\mathcal{O} \in \pi(P)} [\xi_{\text{Pre}_p(\mathcal{O}) \cup \{p\}}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M}) - \xi_{\text{Pre}_p(\mathcal{O})}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})] \quad (24)$$

for $p \in S$. We make a random draw m and compute

$$\begin{aligned} \hat{\theta}_{p,m}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M}) = & \underbrace{\mathcal{M}(\{\sum_{i \in C} [w_{i,t_{j,\text{end}}+1}(\{\hat{r}_{i,t_{j,\text{end}}+1,m,p}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C})r_{i,t_{j,\text{end}}+1}]\}_{j \in W})}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m) \cup \{p\}}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})} - \\ & \underbrace{\mathcal{M}(\{\sum_{i \in C} [w_{i,t_{j,\text{end}}+1}(\{\hat{r}_{i,t_{j,\text{end}}+1,m,\backslash p}(\mathbf{x}_{i,t_{j,\text{end}}}; W_j)\}_{i \in C})r_{i,t_{j,\text{end}}+1}]\}_{j \in W})}_{\hat{\xi}_{\text{Pre}_p(\mathcal{O}_m)}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})}. \end{aligned} \quad (25)$$

The sampling approximation to Equation (24) is then given by

$$\underbrace{\hat{\phi}_p^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})}_{\text{SPPC}_p} = \frac{1}{2M} \sum_{m=1}^{2M} \hat{\theta}_{p,m}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M}) \quad (26)$$

for $p \in S$. Observe that $\hat{\theta}_{p,m}^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})$ in Equation (25) employs Monte Carlo integration to integrate out predictors using background data for the sequence of training samples on which the corresponding sequence of fitted models and concomitant return forecasts is based. Thus, we remain true to each of the fitted models that are used to generate the sequence of return forecasts underpinning portfolio performance.

Again, the properties of Shapley values in Section 2.1 apply to the SPPC_p in Equation (26), including local accuracy, so the contributions of the predictors to the portfolio performance metric sum to the metric (in excess of that for the baseline portfolio):

$$\sum_{p \in S} \underbrace{\hat{\phi}_p^{\text{Perf}}(\{\mathbf{x}_{i,t_{j,\text{end}}}\}_{i \in C}; W, \mathcal{M})}_{\text{SPPC}_p} = \underbrace{\mathcal{M}(\{r_{t_{j,\text{end}}+1}^{\text{Port}}\}_{j \in W})}_{\text{portfolio performance}} - \underbrace{\mathcal{M}(\{r_{t_{j,\text{end}}+1}^{\text{Base}}\}_{j \in W})}_{\text{baseline performance}}. \quad (27)$$

The SPPC_p in Equation (26) measures how an individual predictor contributes to portfolio performance, while Equation (27) indicates that the sum of the SPPC_p measures provides an exact decomposition of portfolio performance relative to the baseline portfolio.

In sum, the Shapley value ascertains the contribution of an individual predictor p to a prediction by forming a coalition of predictors and measuring the change in the prediction when p is included and excluded in the coalition. The change in the value of the prediction corresponding to p is averaged over many coalitions to arrive at the Shapley value, which fairly allocates the contribution of p to the fitted model's prediction. We extend the logic of the Shapley value to a performance metric. To accomplish this, we measure the contribution of predictor p to the performance metric by forming a coalition of predictors and computing the change in value of the metric when p is included and excluded in the coalition in the fitted prediction model (or sequence of fitted models). We average the change in value corresponding to p over many coalitions, providing a Shapley-based approach for fairly allocating the contribution of predictor p to portfolio performance.

We emphasize that the SPPC_p is very general. It is model agnostic, so it applies to any fitted prediction model (linear or nonlinear, parametric or nonparametric). It also

accommodates any rule for mapping the return forecasts to portfolio weights as well as any performance metric.

2.3. Computation

Given the nature of Shapley values, and with more than a few predictors and panel data, estimating the $SPPC_p$ can be computationally costly. Section A1 of the Internet Appendix provides details on implementing our computationally efficient algorithm for calculating the $SPPC_p$.

An alternative approach for measuring the contribution of a predictor is to break the connection between a predictor and the prediction made by a fitted model (or predictions made by a sequence of fitted models). For example, in the spirit of permutation feature importance (Breiman 2001), we replace the predictor value for a given firm and time period with a randomly drawn value from the training sample. (For all of the remaining predictors, we use their original sample values.) We compare the performance metric corresponding to the permuted predictor values to the original performance metric that uses the predictor’s actual sample values, and the difference between the two furnishes an estimate of the predictor’s contribution to portfolio performance.¹¹

Although alternative approaches like those based on predictor permutations are less computationally costly, unlike the $SPPC_p$, they are not based on the logic of Shapley values and thus do not share their attractive properties. For example, the $SPPC_p$ is characterized by local accuracy in Equation (27), so it provides an exact decomposition of any portfolio performance metric with respect to the set of predictors, while predictor permutation does not.¹²

¹¹In the context of developing a framework for constructing an “implementable efficient frontier” via machine learning, Jensen et al. (2026) use this approach to develop an “economic feature importance” measure of how a feature (i.e., predictor) affects realized utility.

¹²Cong et al. (2022) develop a convenient “economic distillation” approach based on a polynomial approximation to measure the importance of predictors for portfolio performance. Like predictor permutation, the approach does not have the properties of Shapley values, including local accuracy.

3. Empirical Application

In this section, we illustrate the use of the $SPPC_p$ developed in Section 2 to analyze a leading question in empirical asset pricing: Which types of firm characteristics are important for determining cross-sectional expected stock returns? In line with recent studies, we analyze the relevance of firm characteristics using out-of-sample tests (e.g., Lewellen 2015; Green, Hand, and Zhang 2017; Freyberger, Neuhierl, and Weber 2020; Gu, Kelly, and Xiu 2020; Avramov, Cheng, and Metzker 2023; Han et al. 2024). As in these studies, we forecast one-month-ahead cross-sectional returns using a large number of firm characteristics. The cross-sectional return forecasts then serve as inputs to form a zero-investment long-short portfolio that is used to measure the economic value of cross-sectional return predictability.

3.1. Data

We use data for a large set of firm characteristics from Chen and Zimmermann (2022), which are available at [Open Source Asset Pricing](#). The data are comprised of 207 firm characteristics from the voluminous literature on cross-sectional expected returns. We use data spanning 1960:01 to 2021:12 (744 months). Following Freyberger, Neuhierl, and Weber (2020) and Gu, Kelly, and Xiu (2020), we transform each characteristic each month by cross-sectionally ranking the characteristics and then mapping the ranks into the $[-1, 1]$ interval. Monthly firm-level stock return data are from CRSP. We consider all firms listed on NYSE, AMEX, and NASDAQ with a market value on CRSP at the end of the previous month and a non-missing value for common equity in the firm’s annual financial statement. We compute the excess return for each stock in a given month using the CRSP risk-free return.¹³

Table 1 lists the 207 firm characteristics from Chen and Zimmermann (2022) along with their descriptions from [Open Source Asset Pricing](#). To help make the decompositions easier to interpret (and reduce computational costs), we consolidate the predictors into 20 groups

¹³As in Gu, Kelly, and Xiu (2020), we fill in missing values for a firm characteristic in a given month with the cross-sectional median for the available characteristic observations for that month.

Table 1. Characteristic groups

The table provides groups for 207 firm characteristics from Chen and Zimmermann (2022) used in the empirical application in Section 3. The characteristics are grouped according to 20 economic categories. The characteristic descriptions are from [Open Source Asset Pricing](#); more information on the data and their sources is provided there.

(1) Description	(2) Description	(3) Description
Panel A: Earnings (9)		
Excluded expenses	Earnings consistency	Earnings streak length
Earnings announcement return	Earnings surprise streak	Analyst earnings per share
Decline in analyst coverage	Earnings surprise	Earnings-to-price ratio
Panel B: Earnings forecast (10)		
Earnings forecast to price	Long-vs-short EPS forecasts	Earnings forecast revisions
EPS forecast revision	Long-term EPS forecast	Up forecast
Δ in forecast and accrual	Predicted analyst forecast error	EPS forecast dispersion
Down forecast EPS		
Panel C: Financing (10)		
Convertible debt indicator	Net equity financing	Leverage component of BM
Δ in current operating liabilities	Net external financing	Market leverage
Δ in financial liabilities	Book leverage (annual)	Net debt to price
Net debt financing		
Panel D: Financing alt (7)		
Composite equity issuance	Initial public offerings	Share issuance (5 year)
Composite debt issuance	Share issuance (1 year)	Spinoffs
Debt issuance		

based on economic concepts.¹⁴ We use the 34 categories in Chen and Zimmermann (2022) as a starting point and make various adjustments to arrive at our 20 groups, with a goal of having groups that are reasonably similar in size. As shown in Table 1, the number of characteristics in a group ranges from six (**Volume**) to 14 (**Investment** and **Profitability**), so no groups are inordinately larger than others. Defining the groups quite narrowly enables us to differentiate between types of characteristics in terms of a wide variety of economic concepts. All of the results for the $SPPC_p$ in Section 2.2 for individual predictors also hold for predictor groups.

¹⁴Harvey, Liu, and Zhu (2016), McLean and Pontiff (2016), Freyberger, Neuhierl, and Weber (2020), and Hou, Xue, and Zhang (2020) categorize characteristics into five to six groups based on economic concepts. We specify a larger number of more narrowly defined economic groups, as we use the $SPPC_p$ to estimate the contributions of the groups to various performance metrics.

Table 1 (continued)

(1) Description	(2) Description	(3) Description
Panel E: Investment (14)		
Cash to assets	Growth in book equity	Investment to revenue
Net operating assets	Δ in equity to assets	Δ in PPE and inv/assets
Real estate holdings	Δ in long-term investment	Growth in advertising exp
Tangibility	Δ in net operating assets	Advertising expense
Asset growth	Growth in long term operating assets	
Panel F: Investment alt (12)		
Δ in capital inv (ind adj)	Inventory growth	Deferred revenue
Δ in capex (2 years)	Δ in net noncurrent op assets	Δ in net financial assets
Δ in capex (3 years)	Δ in net working capital	Employment growth
Brand capital investment	Δ in current operating assets	Total accruals
Panel G: Lead lag (9)		
Customer mom	Customers mom	Price delay coeff
Earnings surprise of big firms	Suppliers mom	Price delay SE adjusted
Industry return of big firms	Price delay R square	Conglomerate return
Panel H: Liquidity (11)		
Pastor-Stambaugh liquidity beta	Probability of informed trading	Days with zero trades (v1)
Bid-ask spread	Size	Days with zero trades (v2)
Amihud's illiquidity	Share turnover volatility	Days with zero trades (v3)
Price		
Panel I: Momentum (11)		
Firm age—mom	Mom (12 month)	Mom in high volume stocks
52 week high	Mom (6 month)	Mom based on FF3 resid
Industry mom	Junk stock mom	Trend factor
Intermediate mom	Mom and LT reversal	
Panel J: Ownership (11)		
Sin stock (selection criteria)	Takeover vulnerability	Inst own and idio vol
Active shareholders	Inst own and forecast dispersion	Short interest
Breadth of ownership	Inst own and market to book	Governance index
Inst own among high short interest	Inst own and turnover	

3.2. Portfolio Construction and Prediction Models

Similarly to a number of recent studies (e.g., Freyberger, Neuhierl, and Weber 2020; Gu, Kelly, and Xiu 2020; Avramov, Cheng, and Metzker 2023; Han et al. 2024), we construct a zero-investment long-short portfolio that goes long (short) stocks with the highest (lowest) machine learning return forecasts for the next month. We use 1960:01 to 1972:12 (156 months) as the initial in-sample estimation period and generate firm-level out-of-sample

Table 1 (continued)

(1) Description	(2) Description	(3) Description
Panel K: Profitability (14)		
Mohanram G-score	Gross profits / total assets	Return on assets (qtrly)
Piotroski F-score	Inventory growth	Net income / book equity
Cash productivity	Operating profits / book equity	Taxable income to income
Cash-based operating profitability	Operating profitability R&D adjusted	O score
Δ in taxes	Operating leverage	
Panel L: R&D (8)		
Citations to R&D expenses	Organizational capital	IPO and no R&D spending
Patents to R&D expenses	R&D over market cap	Unexpected R&D increase
R&D capital-to-assets	R&D ability	
Panel M: Reversal (7)		
Intangible return using BM	Intangible return using Sale2P	Medium-run reversal
Intangible return using CFtoP	Long-run reversal	Short-term reversal
Intangible return using EP		
Panel N: Risk (12)		
Frazzini-Pedersen beta	Coskewness	Idio risk
CAPM beta	Return skewness	Idio risk (3F model)
Tail risk beta	Idio skewness (3F model)	Idio risk (AHT)
Coskewness using daily returns	Systematic volatility	Max return over month
Panel O: Risk alt (12)		
Real dirty surplus	Industry concentration (equity)	IPO and age
Pension funding status	Volatility smirk near the money	Credit Rating Downgrade
Industry concentration (sales)	Put volatility minus call volatility	Exchange Switch
Industry concentration (assets)	Cash-flow to price variance	Firm age based on CRSP

return forecasts and long-short portfolio returns for 1973:01 to 2021:12 (588 months). To keep the fitted prediction model timely when generating the one-month-ahead firm-level return forecasts, we retrain the model each month as additional data become available using a rolling window.

We compute return forecasts using a classification prediction model.¹⁵ There are five classes, from the bottom 20% to top 20% of stocks in terms of their returns in the subsequent month. We describe the classification prediction model in greater detail below, after explaining how we construct the long-short portfolio.

¹⁵As pointed out in Section 2, all of the results for the SPSS_p hold for both regression and classification.

Table 1 (continued)

(1) Description	(2) Description	(3) Description
Panel P: Sales (10)		
Δ in asset turnover	Percent operating accruals	Sales growth / overhead growth
Abnormal accruals	Percent total accruals	Revenue growth rank
Accruals	Sales growth / inventory growth	Order backlog
Δ in order backlog		
Panel Q: Seasonal momentum (10)		
Mom without the seasonal part	Off season reversal years 16–20	Return seasonality years 11–15
Off season long-term reversal	Return seasonality years 2–5	Return seasonality years 16–20
Off season reversal years 6–10	Return seasonality years 6–10	Return seasonality last year
Off season reversal years 11–15		
Panel R: Valuation (12)		
Predicted div yield next month	Dividend seasonality	Δ in recommendation
Efficient frontier index	Share repurchases	Consensus recommendation
Dividend initiation	Equity duration	Analyst recs and short-interest
Dividend omission	Analyst optimism	Analyst value
Panel S: Valuation ratio (11)		
Book-to-market and accruals	Cash flow to market	Sales-to-price
Total assets to market	Enterprise component of BM	Net payout yield
Book to market most recent ME	Enterprise multiple	Payout yield
Book to market December ME	Sales-to-price	
Panel T: Volume (6)		
Past trading volume	Option volume to average	Volume to market equity
Option to stock volume	Share volume	Volume trend

To construct the long-short portfolio for month $t+1$ based on information through month t , we proceed as follows. We generate return forecasts for all available stocks for month $t+1$ using data through month t . Before forming the portfolio, to limit the role of small-cap stocks, we drop stocks with market capitalization below the NYSE 20th percentile at the end of month t . We take long (short) positions in those stocks predicted to be in the top (bottom) quintile of returns in month $t+1$.¹⁶ To further limit the role of small-cap stocks, the long and short legs are value-weighted. We scale the weights in the long and short legs to sum to 1 and -1 , respectively.

We generate monthly firm-level out-of-sample return forecasts based on the 207 firm characteristics listed in Table 1 using the XGBoost algorithm (Chen and Guestrin 2016).

¹⁶Note that the number of stocks in the long leg does not necessarily equal the number in the short leg.

XGBoost is based on a gradient boosting decision tree, which allows for nonlinearities in predictive relations via multiway interactions and higher-order effects of predictors. A decision tree partitions the predictor space into non-overlapping regions and assigns a prediction for the target (i.e., predictand) in each region. The classification and regression tree (CART) algorithm (Breiman et al. 1984) is typically used to partition the predictor space by applying a sequence of splitting rules. The split at the top of a tree is the “root node,” subsequent splits are “internal nodes,” and the final set of subgroups that define the predictive regions at the bottom of the tree are the “terminal” or “leaf nodes.” For a classification tree, the probability prediction for each class is the proportion of target observations in each class in a given leaf node; the predicted class is then the class with the highest proportion of target observations in the leaf node.

XGBoost employs gradient boosting (Breiman 1997; Friedman 2001), which entails constructing an ensemble prediction function additively, where each function in the sequence is a relatively simple model; in the case of a decision tree, each function is a “shallow” tree. Simple models typically have low variance but relatively high bias. Gradient boosting seeks to lower the bias and thus improve out-of-sample performance in light of the bias-variance trade-off by fitting a decision tree to the residuals for the previous tree in the sequence. To help guard against overfitting and further improve out-of-sample performance, stochastic gradient boosting (Friedman 2002) refines conventional gradient boosting by using a randomly drawn subsample of the training data when fitting each decision tree in the sequence. XGBoost is a popular and powerful algorithm that employs stochastic gradient boosting to fit prediction models based on decision trees.¹⁷ It is a leading performer in forecasting competitions and compares favorably to other popular machine learning methods.

An important step for improving out-of-sample performance is tuning the hyperparameters for the XGBoost algorithm.¹⁸ We use a “walk-forward” procedure that respects the

¹⁷We use log loss (or cross entropy) as the objective function when training the XGBoost classification model.

¹⁸We tune the following XGBoost hyperparameters: “max_depth,” “reg_alpha,” “reg_lambda,” “subsample,” “colsample_bytree,” “min_child_weight”; we set “n_estimators” to 100. See the [documentation](#) on

time-series dimension of the panel data to tune the hyperparameters. When computing the first set of out-of-sample return forecasts for 1973:01, we reserve the last 36 months (1970:01 to 1972:12) of the 1960:01 to 1972:12 initial in-sample estimation sample as a validation period for tuning the hyperparameters. We first train prediction models via XGBoost using data for 1960:01 to 1969:12 and the different combinations of hyperparameter values. We plug the characteristic values for 1969:12 into the fitted models to generate return forecasts for 1970:01 and compute long-short portfolio weights as well as the associated portfolio return corresponding to the different combinations of hyperparameter values. Next, we train prediction models via XGBoost using data for 1960:01 to 1970:01 and the different combinations of hyperparameter values, plug the characteristic values for 1970:01 into the fitted models to generate return forecasts for 1970:02, and compute long-short portfolio weights and the associated portfolio return corresponding to the different combinations of hyperparameter values. We continue in this manner through the end of the validation period and compute Sharpe ratios over the validation period for the long-short portfolio returns corresponding to the different combinations of hyperparameter values. We select the combination of hyperparameter values that produces the highest Sharpe ratio over the validation period. Then, using the selected combination of hyperparameter values and 1960:01 to 1972:12 sample, we train the prediction model via XGBoost, plug the 1972:12 characteristic values into the fitted model, and generate the set of return forecasts for the first out-of-sample month (1973:01).

To generate the next set of out-of-sample return forecasts for 1973:02, we use 1960:02 to 1973:01 as the estimation period (so we use a rolling estimation window). We tune the hyperparameters using the walk-forward procedure, with the last 36 months of the estimation period serving as the validation sample. Using the tuned hyperparameters and the

“XGBoost Parameters” for details on the hyperparameters. Because we tune a large number of hyperparameters and consider a wide grid of values for each, it is extremely computationally costly to consider all possible combinations of hyperparameter values (due to the curse of dimensionality). Instead of an exhaustive search, we use **Optuna** (Akiba et al. 2019), which employs the Tree-structured Parzen Estimator (TPE) algorithm to conduct a smart search, thereby substantially reducing computational cost while still making it likely that the selected combination of hyperparameter values is nearly optimal.

1960:02 to 1973:01 estimation sample, we train the prediction model via XGBoost, plug the 1973:01 characteristic values into the fitted model, and produce the set of return forecasts for 1973:02. Continuing in this fashion, we generate firm-level return forecasts based on the 207 characteristics and XGBoost for each month of the forecast evaluation period. The firm-level monthly return forecasts serve as inputs for constructing the long-short portfolio, as described above. The return forecasts and long-short portfolio weights are based on information available at the time of forecast formation so that there is no “look-ahead” bias in the long-short portfolio returns. As shown in Section 3.3, the long-short portfolio performs very well in our empirical application, so the XGBoost algorithm provides an informative machine learning device for analyzing the contributions of cross-sectional stock return predictors to portfolio performance.

3.3. Portfolio Performance and its Decomposition

Table 2 reports performance metrics for the long-short portfolio based on the XGBoost return forecasts for the 1973:01 to 2021:12 forecast evaluation period. For reference, performance metrics are also reported for the CRSP value-weighted aggregate market portfolio, which serves as the baseline. The long-short XGBoost portfolio delivers impressive performance. Its annualized mean return is 22.58%, while its annualized volatility is 12.53%, which compare to values of 7.48% and 15.85%, respectively, for the market portfolio excess return. The mean and volatility for the XGBoost portfolio translate into an annualized Sharpe ratio of 1.80, which is nearly four times larger than that for the market portfolio (0.47). The maximum drawdown for the XGBoost portfolio is only 15.70%, leading to an annualized Calmar ratio of 1.44. Again, the maximum drawdown and Calmar ratio compare favorably to those for the market portfolio (54.11% and 0.14, respectively).

To examine if the XGBoost portfolio generates significant risk-adjusted returns, we estimate its alphas in the context of two leading multifactor models. The first is the Fama and French (2018) six-factor model (FF6), which is comprised of the five Fama and French (2015)

Table 2. Portfolio performance

The table reports performance metrics for a zero-investment long-short portfolio that invests in available stocks in a given month. The long-short portfolio is constructed by sorting stocks according to their excess return forecasts for the available stocks in a given month based on the XGBoost classification model. The excess return forecasts are based on the 207 firm characteristics listed in Table 1. Before forming the portfolio, stocks with market capitalization below the NYSE 20th percentile are dropped. The XGBoost classification portfolio goes long (short) stocks predicted to be in the top (bottom) quintile of excess returns. The long and short legs are value-weighted. The forecast evaluation period is 1973:01 to 2021:12. “MDD” is the maximum drawdown. “Ann. FF6 alpha” is the annualized alpha for Fama and French (2018) six-factor model; “Ann. Q5 alpha” is the annualized alpha for the Hou et al. (2021) augmented q-factor model; t -statistics for the alphas are in brackets; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. “Market” is the CRSP value-weighted market portfolio.

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Portfolio	Ann. mean	Ann. volatility	Ann. Sharpe ratio	MDD	Ann. Calmar ratio	Ann. FF6 alpha	Ann. Q5 alpha
XGBoost	22.58%	12.53%	1.80	15.70%	1.44	19.44% [9.83]***	16.29% [6.93]***
Market	7.48%	15.85%	0.47	54.11%	0.14	0%	0%

factors and a momentum factor. The second is the Hou et al. (2021) augmented q-factor model (Q5), which adds an expected investment growth factor to the four factors from the original q-factor model (Hou, Xue, and Zhang 2015).¹⁹ The XGBoost portfolio generates economically sizable annualized alphas of 19.44% and 16.29% for the FF6 and Q5 multifactor models, respectively. Both alpha estimates are significant at the 1% level. This indicates that exposures to popular systematic risk factors from the literature cannot account for the average return of the XGBoost portfolio. Indeed, the risk-adjusted average returns (i.e., alphas) for the XGBoost portfolio in the last two columns of Table 2 are reasonably close to its unadjusted average return in the second column.²⁰

¹⁹The factor data for the FF6 (augmented q-factor) model are from Kenneth French’s [Data Library](#) (Lu Zhang’s [website](#)).

²⁰A long-short portfolio based on return forecasts generated by an XGBoost regression model also outperforms the market portfolio in Table 2. However, it does not perform as well as the long-short portfolio based on the XGBoost classification model. It is perhaps not surprising that the XGBoost classification model works better for constructing the long-short portfolio than the XGBoost regression model, as forming the long-short portfolio essentially entails classification, with the portfolio going long (short) stocks that are predicted to be in the top (bottom) quintile in terms of their returns in the next month. In some situations,

Overall, the results in Table 2 demonstrate that the XGBoost return forecasts based on the 207 firm characteristics serve as valuable inputs for constructing a long-short portfolio. In what follows, we employ the $SPPC_p$ in Equation (26) to measure the contributions of each of the 20 predictor groups in Table 1 to the performance metrics for the XGBoost portfolio. While we focus on decomposing the performance metrics for the XGBoost portfolio, we reiterate that the $SPPC_p$ can be used to decompose the performance of any portfolio constructed from return forecasts based on any model and set of predictors. If a portfolio performs poorly, our method can identify the predictors that are primarily responsible for the subpar performance.

As discussed in Section 2.2, to operationalize our method, we need to select the return for the portfolio corresponding to the empty coalition set, which serves as a baseline. If an investor does not have access to any predictors (the empty coalition set), it seems reasonable to simply hold the market portfolio; thus, we use the excess return for the CRSP value-weighted market portfolio as the baseline. Observe that, because the market excess return appears as a factor in both the FF6 and Q5 multifactor models, the alpha for the baseline portfolio is zero by construction.

Table 3 reports the contributions of the 20 predictor groups to the performance metrics for the XGBoost portfolio. The top four contributions for each metric in terms of improving portfolio performance are in bold. According to the local accuracy property in Equation (27), the baseline contribution and those of the 20 predictor groups sum to the total in the last row of Table 3 (apart from rounding), where the last row equals the corresponding value for the XGBoost portfolio in Table 2.

Beginning with the mean return in Table 3, the baseline value for the market portfolio is 7.48%. Since the mean return for the XGBoost portfolio is 22.58%, the 20 predictor groups together increase the average return by 15.10 percentage points. The top four predictor

we need return forecasts based on a regression model (e.g., to construct mean-variance optimal portfolios), and the $SPPC_p$ can be used to measure the contributions of the predictors to portfolio performance in this context (see Section 2).

Table 3. Portfolio performance contributions

The table reports the contributions of the 20 predictor groups to the performance metrics in Table 2 for the long-short XGBoost classification portfolio. Table 1 lists the individual firm characteristics in each of the 20 predictor groups. The forecast evaluation period is 1973:01 to 2021:12. The numbers in a column may not add to the value in the “Total” row due to rounding; 0.00 indicates less than 0.005 in absolute value. The top four contributions for each metric in terms of improving portfolio performance are in bold.

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Predictor group	Ann. mean	Ann. vol.	Ann. Sharpe ratio	MDD	Ann. Calmar ratio	Ann. FF6 alpha	Ann. Q5 alpha
Baseline	7.48%	15.85%	0.47	54.11%	0.14	0%	0%
Risk	5.01	−0.09	0.38	−4.41	0.26	4.67	4.59
Earnings	2.67	−0.21	0.20	−4.16	0.12	2.82	2.18
Seasonal momentum	1.72	−0.91	0.17	−7.69	0.16	2.53	2.65
Momentum	4.38	2.24	0.14	3.88	0.08	3.18	1.85
Lead lag	0.90	−0.75	0.09	−2.31	0.04	0.82	0.40
Investment	0.96	−0.45	0.09	−5.60	0.10	1.58	0.75
Valuation ratio	0.00	−1.01	0.08	−5.77	0.13	−0.09	0.16
Risk alt	0.42	−0.66	0.08	−5.89	0.17	1.19	0.71
Earnings forecast	0.92	0.05	0.06	−0.43	0.03	0.86	1.01
Valuation	0.24	−0.78	0.06	−5.23	0.07	0.57	0.38
Profitability	1.00	0.08	0.05	−3.04	0.08	0.60	−0.88
Liquidity	0.28	0.98	0.04	0.98	0.07	0.21	0.75
Financing	−0.04	0.07	0.00	−0.14	0.00	0.02	0.24
Financing alt	0.05	0.10	0.00	0.37	0.02	0.71	0.24
Volume	−0.87	−1.19	0.00	−2.09	0.01	−0.45	0.16
Investment alt	−0.04	0.06	0.00	0.15	0.01	0.11	−0.04
R&D	−0.04	0.16	−0.01	−1.08	0.01	0.62	0.18
Sales	−0.35	−0.19	−0.02	0.61	0.00	−0.01	−0.36
Ownership	−0.61	−0.60	−0.03	−0.46	−0.02	−0.16	−0.17
Reversal	−1.50	−0.20	−0.04	3.90	−0.04	−0.34	1.50
Total	22.58%	12.53%	1.80	15.70%	1.44	19.44%	16.29%

groups are Risk, Momentum, Earnings, and Seasonal momentum, with contributions of 501, 438, 267, and 172 basis points, respectively. These four groups collectively contribute to an increase in the average return of 13.78 percentage points, which is over 90% of the increase provided by the XGBoost portfolio. Other groups making contributions above 50 basis points are Profitability, Investment, Earnings forecast, and Lead lag (100, 96, 92, and 90

basis points, respectively). A handful of groups make sizably negative contributions to the average return, including **Reversal**, **Volume**, **Ownership**, and **Sales** (-150 , -87 , -61 , and -35 basis points, respectively). With respect to volatility, the XGBoost portfolio lowers it by 3.32 percentage points vis-à-vis the market portfolio. The largest contributors to the reduction in volatility are **Volume**, **Valuation ratio**, **Seasonal momentum**, and **Valuation** (-119 , -101 , -191 , and -78 basis points, respectively).

In terms of the Sharpe ratio in Table 3, the 20 predictor groups collectively increase the Sharpe ratio by a considerable 1.33 for the XGBoost portfolio relative to the market portfolio (0.47 to 1.80). **Risk**, **Earnings**, **Seasonal momentum**, and **Momentum** are the top four predictor groups, with contributions of 0.38, 0.20, 0.17, and 0.14, respectively. Together, these four groups contribute to an increase in the Sharpe ratio of 0.89 or 67% of the total increase. The remaining predictor groups can be divided into three categories. First, there are groups making smaller but still noteworthy contributions in the range of 0.04 to 0.09, led by **Lead lag** and **Investment**, both with contributions of 0.09. Second, a set of groups makes minor contributions of essentially zero. Third, four groups that contribute negatively to the average return—**R&D**, **Sales**, **Ownership**, and **Reversal**—also make negative contributions to the Sharpe ratio (-0.01 , -0.02 , -0.03 , and -0.04 , respectively).

The XGBoost portfolio reduces the maximum drawdown for the market portfolio from 54.11% to 15.70%, a substantive reduction of 38.41 percentage points. The groups most responsible for the decrease include **Seasonal momentum**, **Risk alt**, **Valuation ratio**, and **Investment**, with contributions of -769 , -589 , -577 , and -560 basis points, respectively. Other groups lowering the maximum drawdown by more than three percentage points are **Valuation**, **Risk**, **Earnings**, and **Profitability** (-523 , -441 , -416 , and -304 basis points, respectively).

The results for the Calmar ratio are broadly similar to those for the Sharpe ratio. The XGBoost portfolio increases the Calmar ratio from 0.14 for the market portfolio to 1.44, a substantial increase of 1.30. The **Risk** group makes the largest contribution (0.26) to

the increase in the Calmar ratio. In addition to **Risk**, groups making contributions of 0.10 or more include **Risk alt**, **Seasonal momentum**, **Valuation ratio**, **Earnings**, and **Investment** (0.17, 0.16, 0.13, 0.12, and 0.10, respectively). As in the case for the Sharpe ratio, **Ownership** and **Reversal** make negative contributions (-0.02 , -0.04 , respectively) to the Calmar ratio.

Turning to the alphas for the FF6 and Q5 multifactor models in Table 3, the top four predictor groups in terms of contributions to the Sharpe ratio also comprise the top four in terms of contributions to the FF6 and Q5 alphas. For the FF6 model, **Risk**, **Momentum**, **Earnings**, and **Seasonal momentum** contribute 467, 318, 282, and 253 basis points, respectively, to alpha; together, these groups account for 13.20 percentage points of the total alpha for the XGBoost portfolio of 19.44%. Other groups contributing more than 50 basis points include **Investment**, **Risk alt**, **Earnings forecast**, **Lead lag**, **Financing alt**, **R&D**, **Profitability**, and **Valuation** (158, 119, 86, 82, 71, 62, 60, and 57 basis points, respectively). **Ownership** and **Reversal** make sizable negative contributions (-16 and -34 basis points, respectively).

For the Q5 model, **Risk**, **Seasonal momentum**, **Earnings**, and **Momentum** provide contributions of 459, 265, 218, and 185 basis points, respectively, to alpha. These contributions constitute 11.27 percentage points of the total alpha of 16.29%. Other noteworthy contributions in excess of 50 basis points are made by **Earnings forecast**, **Investment**, **Liquidity**, and **Risk alt** (101, 75, 75, 71 basis points, respectively). **Profitability**, **Sales**, **Ownership**, and **Investment alt** make negative contributions (-88 , -36 , -17 , and -4 basis points, respectively).

Overall, the $SPPC_p$ values reported in Table 3 enable us to identify the predictor groups that are primarily responsible for the strong performance of the XGBoost portfolio according to the metrics reported in Table 2. Four groups—**Risk**, **Earnings**, **Seasonal momentum**, and **Momentum**—stand out in Table 3 as leading contributors to the economic value provided by cross-sectional return predictability. These groups make the largest contributions to the

Sharpe ratio as well as the alphas for the FF6 and Q5 multifactor models. There are also groups that often substantially detract from portfolio performance, especially **Sales**, **Ownership**, and **Reversal**.

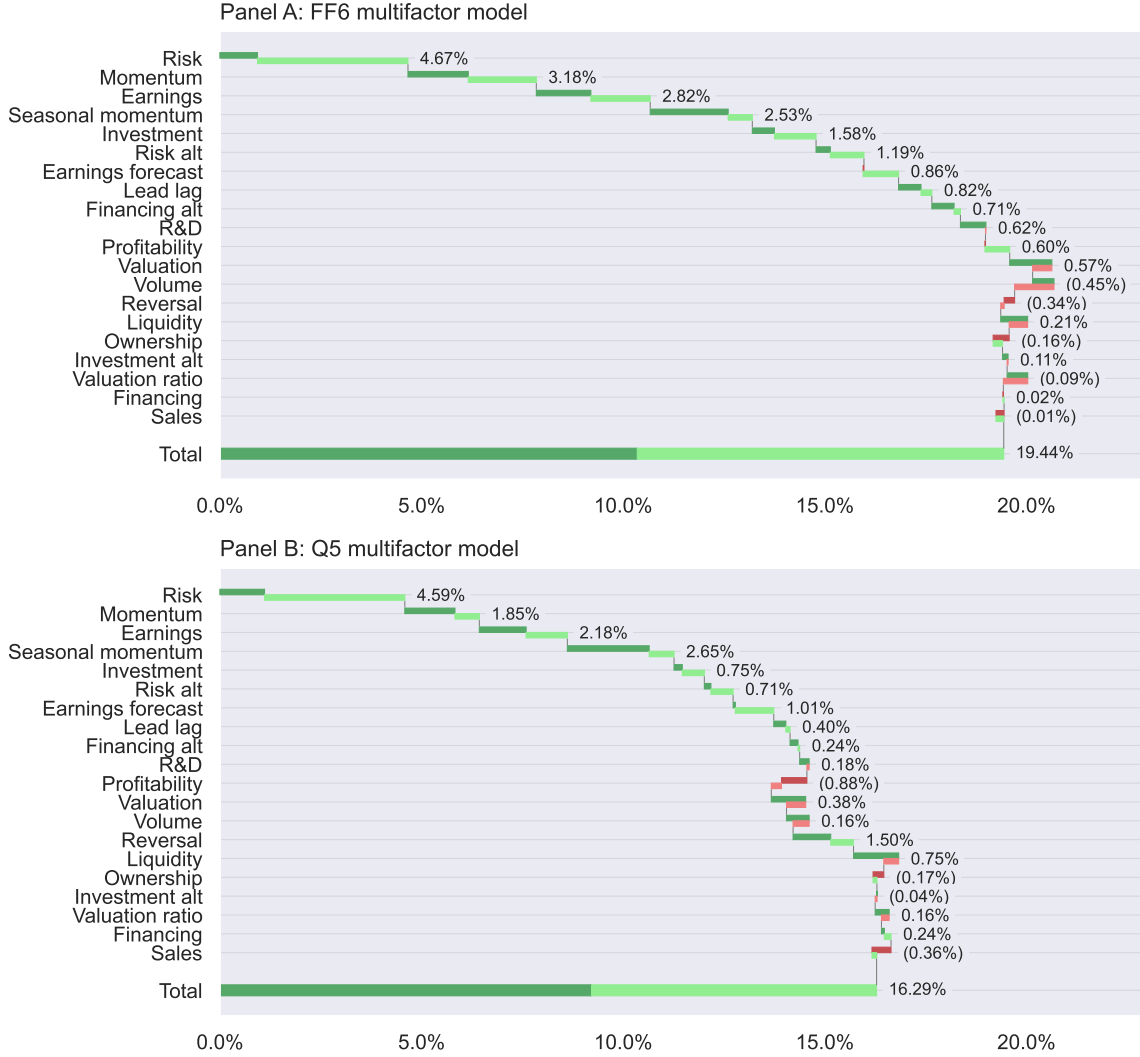


Figure 1. Alpha long- and short-leg contributions. Each panel depicts a waterfall diagram with the contributions of the 20 predictor groups to the annualized alpha for the long-short XGBoost classification portfolio in terms of the long and short legs. The dark (light) green segments are the positive contributions to the risk-adjusted return for the long (short) leg; the dark (light) red segments are the negative contributions to the risk-adjusted return for the long (short) leg. The numbers correspond to the contributions of the long and short legs together reported in the last two columns of Table 3; parentheses indicate a negative number.

Researchers are often interested in how the long and short legs separately affect the performance of long-short portfolios. We can use the $SPPC_p$ to address this issue. As an

example, Figure 1 presents waterfall diagrams depicting the contributions of the predictor groups to the FF6 and Q5 multifactor model alphas for the long-short XGBoost portfolio in terms of the long and short legs separately. The diagram includes the group contributions to the alphas reported in the last two columns of Table 3 (with negative numbers in parentheses). Panel A (B) of Figure 1 reports results for the FF6 (Q5) model. The dark (light) green segments are the positive contributions to the risk-adjusted return for the long (short) leg;²¹ the dark (light) red segments are the negative contributions to the risk-adjusted return for the long (short) leg. The base of the waterfall shows the total contributions of the long and short legs to the alpha. Overall, for both multifactor models, the long and short legs contribute approximately equally to the long-short portfolio alpha.

There are interesting contrasts in the contributions of some of the predictor groups across the long and short legs. Consider, for example, **Risk**, which makes the largest contribution to the long-short portfolio alpha for both multifactor models. In both cases, the contribution of **Risk** to the alpha is considerably larger for the short leg. Other groups whose short-leg contributions are substantially larger than their long-leg contributions for both models include **Investment** and **Earnings forecast**. In contrast, the long-leg contributions are relatively large for **Seasonal momentum**. For **Momentum** and **Earnings**, two of the leading predictor groups for both models, the long- and short-leg contributions are fairly similar.

3.4. Time Variation in Predictor Contributions

To this point, we have computed predictor group contributions for portfolio performance metrics over the full 1973:01 to 2021:12 forecast evaluation period. The $SPPC_p$ can be used to measure predictor contributions for any subsample, thereby allowing for the investigation of time variation in the contributions. To provide motivation for examining time variation in predictor contributions, Figure 2 shows the cumulative log return for the long-short XGBoost portfolio for the 1973:01 to 2021:12 out-of-sample period. For reference, the cumulative log

²¹We use the negative of the return for the short leg, as this represents a positive contribution to the long-short portfolio alpha.

excess return for the CRSP value-weighted market portfolio is also shown. The XGBoost portfolio performs especially well through the early 2000s, after which its cumulative log return “flattens out” to an extent.²² The XGBoost portfolio performs considerably better than the market portfolio during business-cycle recessions throughout the entire 1973:01 to 2021:12 forecast evaluation period. After the early 2000s, the XGboost portfolio continues to perform particularly well during recessions—especially the Great Recession—but its performance is less impressive during expansions compared to earlier. Investor learning about cross-sectional return predictability from academic studies (McLean and Pontiff 2016) potentially provides a partial explanation for the decrease in performance for the XGBoost portfolio after the early 2000s.

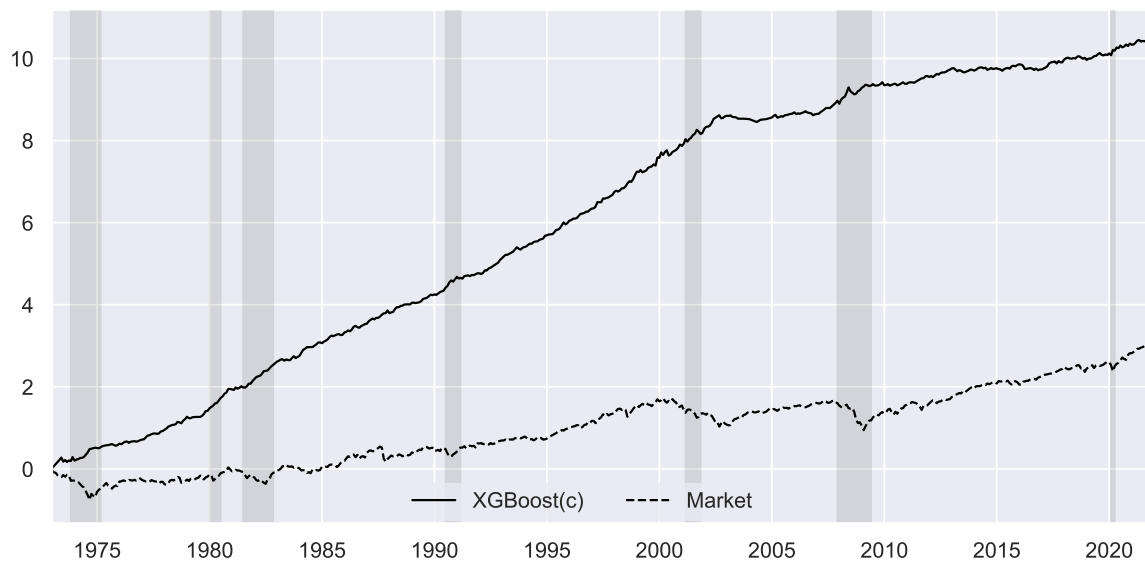


Figure 2. Cumulative log returns. The figure depicts the cumulative log return for the long-short XGBoost classification (XGBoost(c)) portfolio and the cumulative log excess return for the CRSP value-weighted market portfolio. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

²²Green, Hand, and Zhang (2017) obtain a similar finding for a long-short portfolio constructed from sorted return forecasts generated using linear Fama and MacBeth (1973) regressions estimated via ordinary least squares based on 94 firm characteristics. A similar pattern is also evident in Han et al. (2024), who construct a long-short portfolio from sorted return forecasts generated using Fama-MacBeth regressions estimated with machine learning methods based on 207 firm characteristics.

We investigate changes in portfolio performance and the predictor group contributions over time via rolling windows of data over the forecast evaluation period.²³ We compute performance metrics for the long-short XGBoost portfolio based on 60-month rolling windows and then estimate the predictor group contributions via the $SPPC_p$ for the rolling windows. Figure 3 displays the sequences of Sharpe and Calmar ratios as well as FF6 and Q5 alphas computed using the rolling windows. The performance metrics tend to fall when data beyond the early 2000s are included in the window. In addition, there is a tendency for the metrics to increase when the windows incorporate data from recessions. This is clearly evident for the Sharpe ratio and the alphas around the Great Recession. The performance metrics also markedly increase for windows that include data near the end of the sample, corresponding to the advent of the COVID-19 crisis.

Figures 4 and 5 depict the contributions of the 20 predictor groups to the Sharpe and Calmar ratios, respectively, computed for the 60-month rolling windows. For a given window, we standardize the contributions by the maximum group contribution for that window.²⁴ The results in Figures 4 and 5 are similar and reveal noteworthy patterns in the group contributions over time. For example, **Risk** and **Momentum** frequently make among the largest contributions for windows ending through the early 2000s, while they subsequently often make sizably negative contributions. Other groups making contributions that substantially vary between positive and negative values include **Profitability**, **Earnings forecast**, and **Reversal**. Groups making more consistent positive contributions over time include **Earnings**, **Seasonal momentum**, and **Investment**. In general, the contributions in Figures 4 and 5 appear more stable for windows ending through the early 2000s.

Figures 6 and 7 show the contributions of the predictor groups to the alphas for the FF6 and Q5 multifactor models, respectively, computed for the 60-month rolling windows. The

²³Section A2 of the Internet Appendix reports portfolio performance metrics and predictor group contributions for the 1973:01 to 2002:12 and 2003:01 to 2021:12 subsamples.

²⁴Thus, a line in Figure 4 has a maximum value of one when the group’s contribution is equal to the maximum contribution for that window; a standardized contribution of -1 means that the group makes a negative contribution that is equal in magnitude to the maximum contribution for that window.

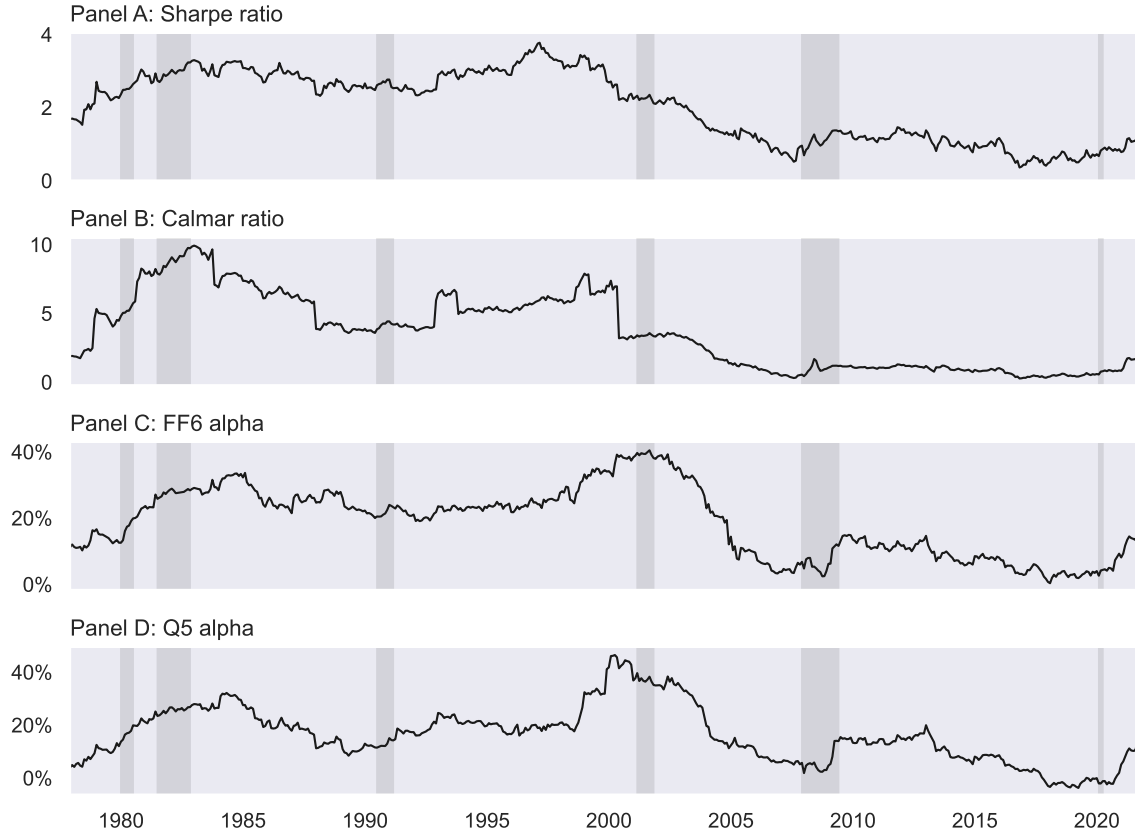


Figure 3. Portfolio performance for 60-month rolling windows. Each panel depicts the annualized performance metric in the panel heading for the long-short XGBoost classification portfolio. The metrics are computed using 60-month rolling windows over the 1973:01 to 2021:12 forecast evaluation period. The horizontal axis corresponds to the end of the 60-month rolling window. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

two figures tell a similar story with respect to the contributions to the alphas over time. For windows ending prior to the early 2000s, **Risk** nearly always contributes positively and sizably to the alphas, but it often makes large negative contributions thereafter. **Momentum** and **Reversal** contribute erratically to the alphas, making both substantive positive and negative contributions after the early 2000s. **Earnings**, **Seasonal momentum**, and (to a lesser extent) **Investment** make sizably positive contributions on a reasonably consistent basis over time. Overall, similarly to Figures 4 and 5, the contributions tend to exhibit greater fluctuations in Figures 6 and 7 after the early 2000s.

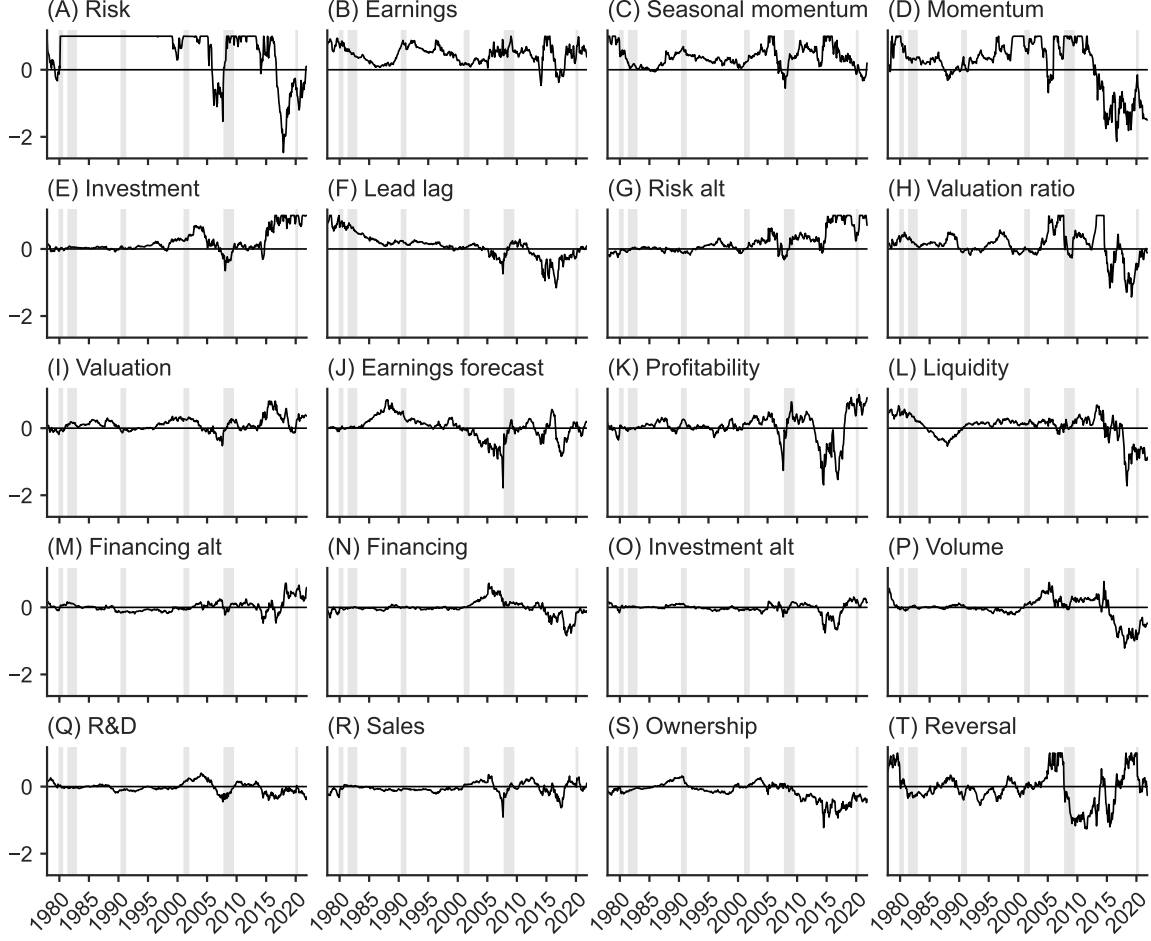


Figure 4. Sharpe ratio contributions for 60-month rolling windows. The figure depicts the contributions of the 20 predictor groups to annualized Sharpe ratios for the long-short XGBoost classification portfolio. The Sharpe ratios are computed using 60-month rolling windows over the 1973:01 to 2021:12 forecast evaluation period. The horizontal axis corresponds to the end of the 60-month rolling window. The contributions are standardized by the maximum contribution to the Sharpe ratio in a given rolling window. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

3.5. Alternative Approach

As a final exercise, we decompose the alphas in Table 2 using the predictor permutation method described in Section 2.3. It provides an alternative approach for measuring a predictor’s contribution to portfolio performance by breaking the connection between the predictor and fitted model prediction. The permutation method is less computationally costly than the $SPPC_p$. However, as pointed out in Section 2.3, unlike the $SPPC_p$, the permutation method

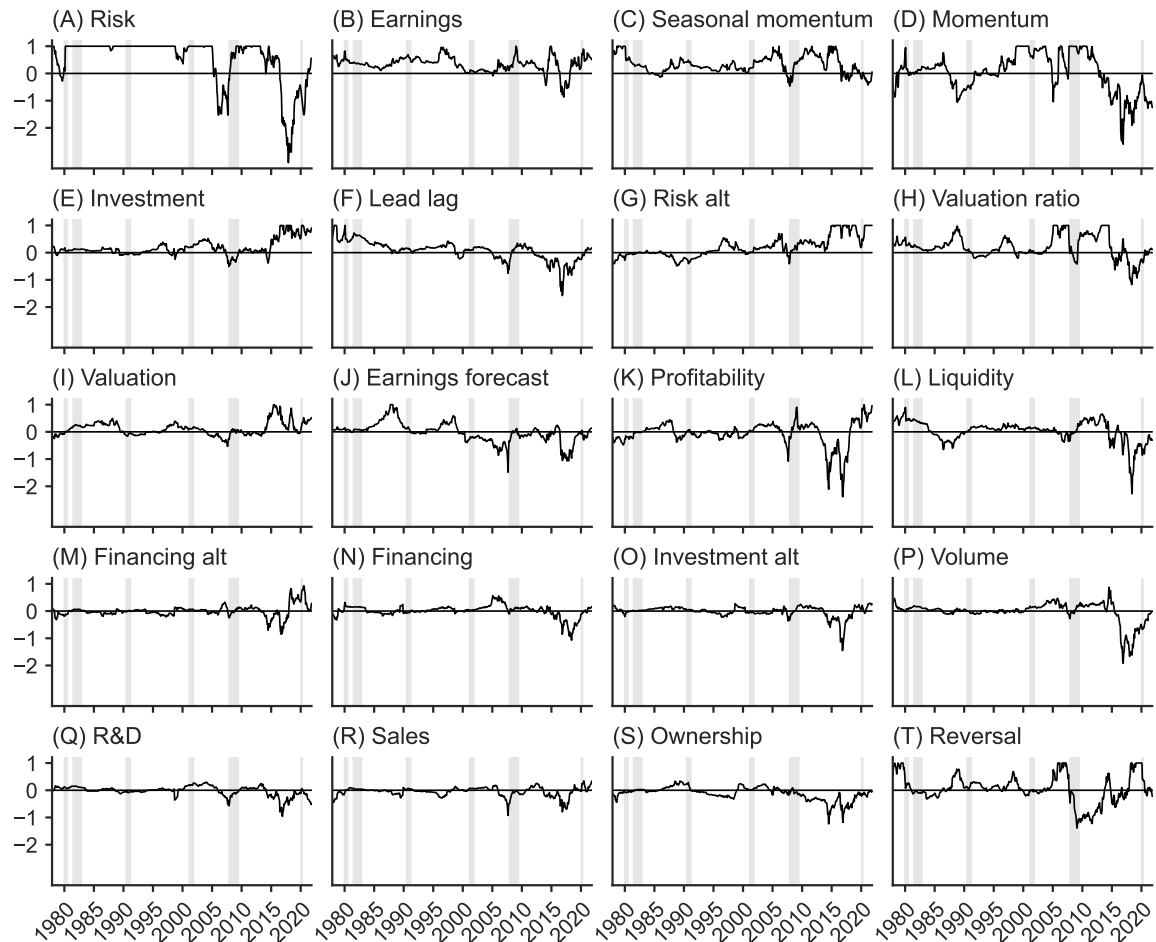


Figure 5. Calmar ratio contributions for 60-month rolling windows. The figure depicts the contributions of the 20 predictor groups to annualized Calmar ratios for the long-short XGBoost classification portfolio. The Calmar ratios are computed using 60-month rolling windows over the 1973:01 to 2021:12 forecast evaluation period. The horizontal axis corresponds to the end of the 60-month rolling window. The contributions are standardized by the maximum contribution to the Calmar ratio in a given rolling window. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

does not possess the attractive properties of Shapley values. We focus on the alphas, as the baseline market portfolio has a value of zero in this case. Because the permutation method does not readily accommodate a baseline like the market portfolio, we cannot directly compare the $SPPC_p$ and permutation decompositions when the performance metric for the baseline is nonzero.²⁵

²⁵This is an advantage of the Shapley-based $SPPC_p$, as researchers are often interested in the performance of an actively managed portfolio relative to a passively managed benchmark.

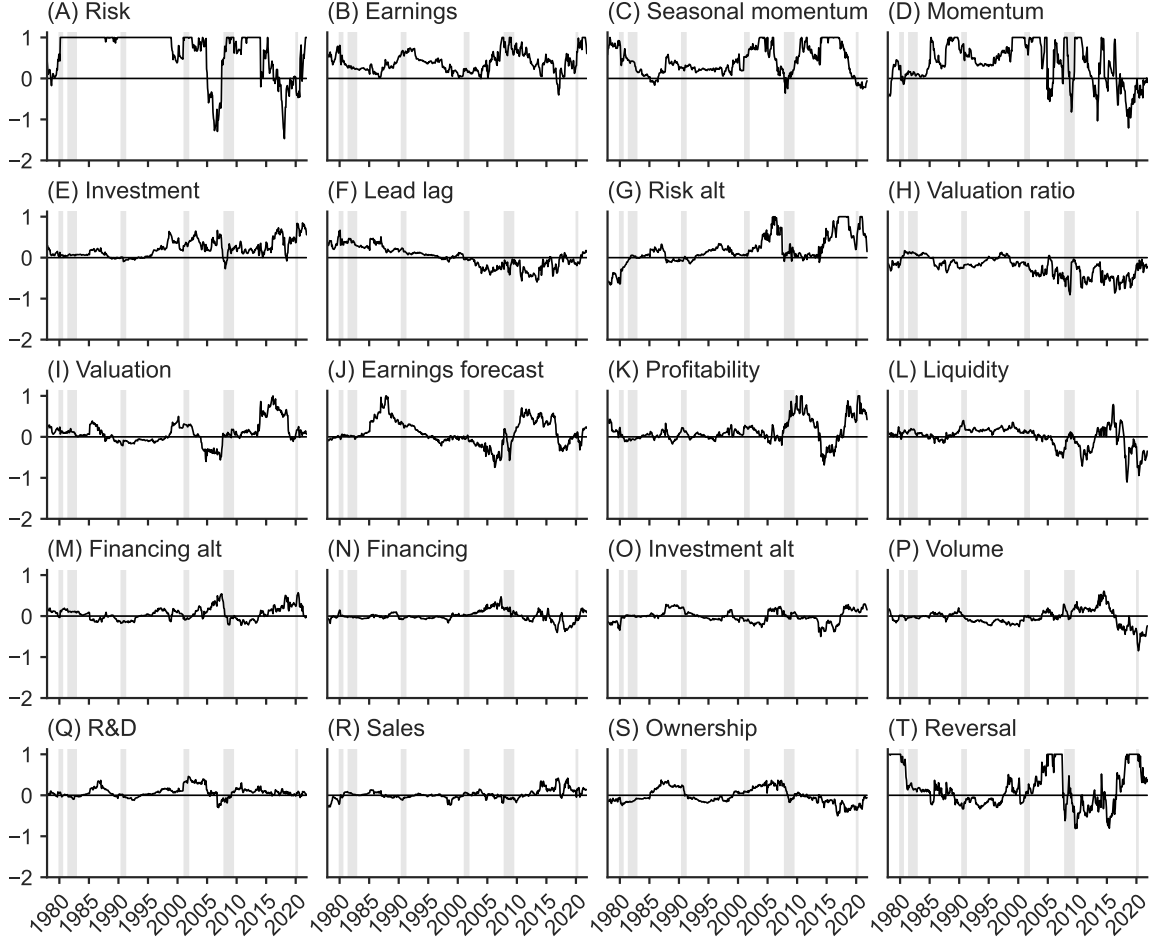


Figure 6. FF6 alpha contributions for 60-month rolling windows. The figure depicts the contributions of the 20 predictor groups to annualized alphas for the long-short XGBoost classification portfolio. The alphas are estimated in the context of the FF6 multifactor model and computed using 60-month rolling windows over the 1973:01 to 2021:12 forecast evaluation period. The horizontal axis corresponds to the end of the 60-month rolling window. The contributions are standardized by the maximum contribution to the alpha in a given rolling window. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

Table 4 reports decompositions of the FF6 and Q5 alphas. For ease of comparison, the second and fourth columns reproduce the $SPPC_p$ decompositions from the last two columns of Table 3, while the third and fifth columns of Table 4 report decompositions using the permutation method. Beginning with the FF6 alpha, we again see from the second column that the $SPPC_p$ provides an exact decomposition of the alpha for the long-short XGBoost portfolio (19.44%). According to the third column, the permutation method does not provide an exact decomposition, as the 20 predictor group contributions sum to 36.20%, which is

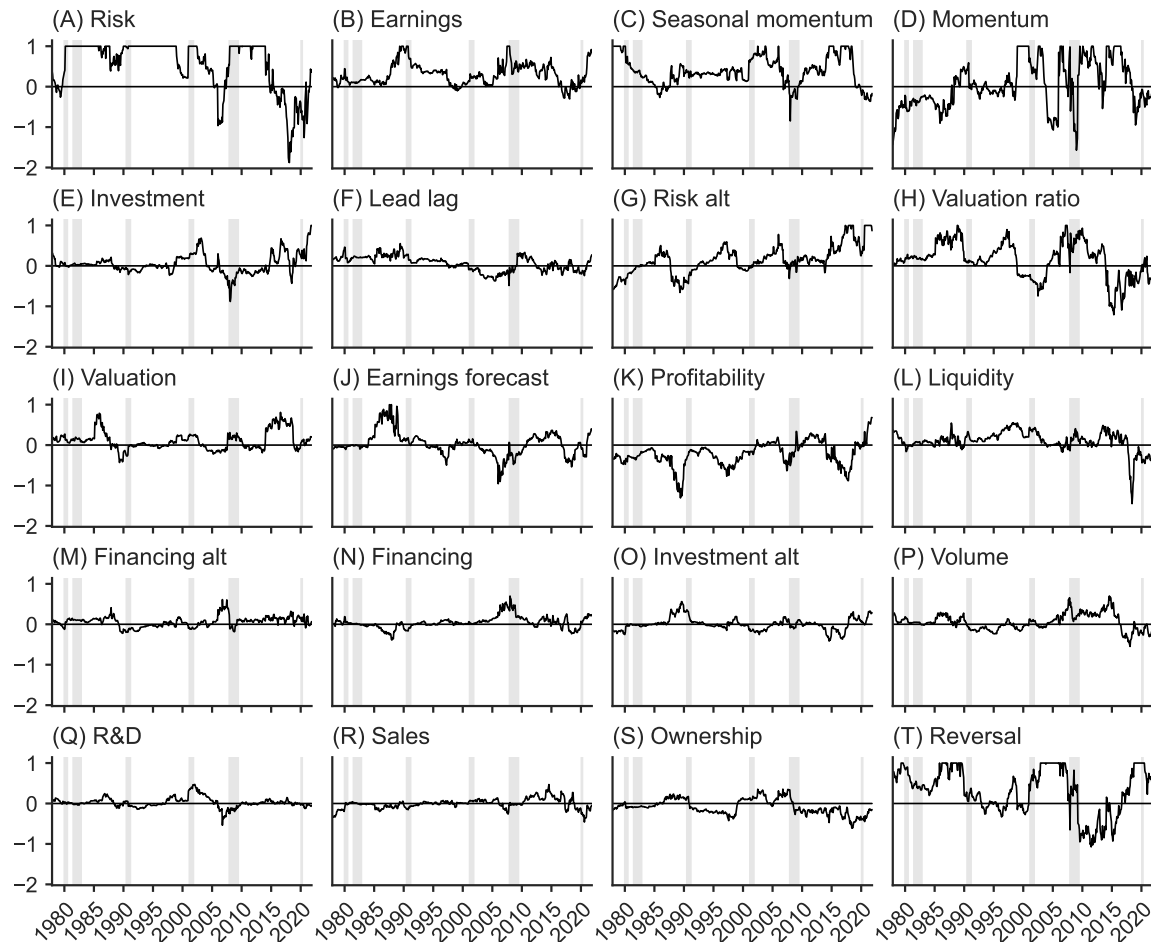


Figure 7. Q5 alpha contributions for 60-month rolling windows. The figure depicts the contributions of the 20 predictor groups to annualized alphas for the long-short XGBoost classification portfolio. The alphas are estimated in the context of the Q5 multifactor model and computed using 60-month rolling windows over the 1973:01 to 2021:12 forecast evaluation period. The horizontal axis corresponds to the end of the 60-month rolling window. The contributions are standardized by the maximum contribution to the alpha in a given rolling window. Vertical bars delineate business-cycle recessions as dated by the National Bureau of Economic Research.

nearly twice as large as the actual FF6 alpha. Furthermore, there are a number of substantive discrepancies between the measured contributions for the two approaches. For example, the signs of the contributions are opposite for five predictor groups (**Valuation ratio**, **R&D**, **Sales**, **Ownership**, and **Reversal**), and the magnitudes of the differences are fairly sizable, ranging from 46 to 150 basis points (**Reversal** and **Valuation ratio**, respectively). The contributions of other predictor groups also differ by sizable magnitudes, such as **Risk**, **Momentum**, and **Earnings** (307, 290, and 200 basis points, respectively); in these cases, the

Table 4. Alpha contributions using an alternative method

The table reports the contributions of the 20 predictor groups to the annualized FF6 and Q5 alphas for the long-short XGBoost classification portfolio. The contributions in the second and fourth columns are reproduced from the seventh and eighth columns of Table 3; the numbers in the second and fourth columns may not add to the value in the “Total” row due to rounding. The contributions in the third and fifth columns are estimated using the alternative (permutation) method. The forecast evaluation period is 1973:01 to 2021:12.

(1) Predictor group	(2) Ann. FF6 alpha		(4) Ann. Q5 alpha	
	SPPC _p	Permutation	SPPC _p	Permutation
Risk	4.67	7.74	4.59	7.31
Earnings	2.82	4.82	2.18	4.00
Seasonal momentum	2.53	1.93	2.65	2.69
Momentum	3.18	6.08	1.85	5.32
Lead lag	0.82	1.83	0.40	0.94
Investment	1.58	2.00	0.75	0.46
Valuation ratio	−0.09	1.41	0.16	1.37
Risk alt	1.19	1.94	0.71	1.55
Earnings forecast	0.86	1.15	1.01	0.60
Valuation	0.57	1.65	0.38	0.73
Profitability	0.60	2.28	−0.88	0.46
Liquidity	0.21	0.42	0.75	1.35
Financing	0.02	1.67	0.24	1.64
Financing alt	0.71	0.59	0.24	−0.36
Volume	−0.45	−1.08	0.16	−0.80
Investment alt	0.11	0.97	−0.04	1.28
R&D	0.62	−0.19	0.18	−0.32
Sales	−0.01	0.56	−0.36	−0.90
Ownership	−0.16	0.31	−0.17	0.23
Reversal	−0.34	0.12	1.50	0.94
Total	19.44%	36.20%	16.29%	28.49%

estimated contributions are higher for the permutation method, helping to explain why the approach overstates the alpha when all of the predictors are taken together.

The results are broadly similar for the Q5 alpha decompositions in the last two columns of Table 4. In contrast to the exact decomposition provided by the SPPC_p, based on the permutation method, the 20 predictor group contributions sum to 28.49%, which is nearly

75% larger than the actual alpha (16.29%). For six of the predictor groups, the signs of the contribution are opposite (**Profitability**, **Financing alt**, **Volume**, **R&D**, **Sales**, and **Ownership**), and the magnitudes of the gaps range from 40 to 134 basis points (**Ownership** and **Profitability**, respectively). Substantive differences in the magnitudes of the estimated contributions are also evident for **Momentum**, **Risk**, and **Earnings** (347, 272, and 182 basis points, respectively).

Overall, we interpret the results in Table 4 as favoring the $SPPC_p$ over the permutation approach for decomposing portfolio performance, even though the latter is less computationally costly. Because the measured contributions often differ substantially, it matters which method is used; since the $SPPC_p$ is based on the logic of Shapley values and possesses their attractive properties, it fairly allocates the contributions across predictors and provides a more informative measure of a predictor’s contribution to portfolio performance, thereby making it the preferred option from a cost-benefit perspective.

4. Conclusion

Asset return predictability is now commonly assessed in terms of economic value. A researcher generates out-of-sample return forecasts for one or more assets, increasingly using a large set of predictors and a machine learning model. The return forecasts then serve as inputs to construct a portfolio over the forecast evaluation period, and performance metrics are used to measure the economic value of return predictability from an investment perspective. While measuring the economic value of return predictability is crucial for assessing the relevance of return predictability, it is also vital to understand the sources of the economic value provided by return predictability.

The information in the underlying predictors in fitted machine learning models is the ultimate source of return predictability and its associated economic value. However, the existing literature does not provide a general procedure for exactly decomposing economic value as measured by a portfolio performance metric into the contributions of the under-

lying predictors. The present paper fills this gap by developing the SPPC_p , a new model interpretation tool for finance founded on Shapley values that measures the contributions of individual or groups of predictors in fitted prediction models to portfolio performance. Based on the logic of Shapley values, the SPPC_p fairly allocates the predictor contributions to the portfolio performance metric. Shapley values have a number of attractive properties inherited by the SPPC_p , including local accuracy, so the SPPC_p values for the set of predictors provide an exact decomposition of the performance metric in terms of the underlying predictors. The SPPC_p is very flexible: it can be used for any prediction model, including any machine learning model; any strategy for mapping return forecasts to portfolio weights; and any performance metric. In sum, the SPPC_p provides an anatomy of machine learning-based portfolio performance and constitutes a powerful tool for deepening our understanding of the sources of the economic value generated by return predictability.

We illustrate the use of the SPPC_p in an empirical application investigating firm-level stock return predictability based on 207 firm characteristics from Chen and Zimmermann (2022). We employ the XGBoost algorithm, a powerful machine learning device, to generate monthly out-of-sample return classification forecasts based on the firm characteristics for 1973:01 to 2021:12, where individual stocks are predicted to be in quintiles in terms of their returns for the next month. Based on the forecasts, we construct a zero-investment portfolio that goes long (short) stocks that are predicted to be in the top (bottom) quintile. To minimize the role of small-cap stocks when forming the portfolio, we drop stocks with market capitalization below the NYSE 20th percentile and employ value-weighting in the long and short legs. The long-short portfolio delivers substantial economic value in terms of Sharpe and Calmar ratios as well as risk-adjusted returns in the context of leading multifactor models.

We categorize the firm characteristics into 20 groups according to economic concepts and estimate the SPPC_p for the predictor groups and portfolio performance metrics. Groups making the largest positive contributions to portfolio performance over the full 1973:01

to 2021:12 forecast evaluation period include **Risk**, **Earnings**, **Seasonal momentum**, and **Momentum**, while **Sales**, **Ownership**, and **Reversal** often make negative contributions. The performance of the long-short portfolio generally declines after the early 2000s, but it still performs relatively well, especially during business-cycle recessions. To shed light on the sources of the change in portfolio performance over time, we estimate the $SPPC_p$ for the predictor groups for rolling windows from the full forecast evaluation period. The $SPPC_p$ measures reveal that the contributions of **Risk** and **Momentum** to the performance metrics typically decrease substantively after the early 2000s, often becoming negative. In contrast, **Earnings**, **Seasonal momentum**, and **Investment** make positive and sizable contributions to portfolio performance on a relatively consistent basis over time, indicating that these characteristic groups are more reliable predictors of cross-sectional stock returns in a machine learning framework when it comes to economic value. Our empirical application demonstrates how the $SPPC_p$ provides insights into the key determinants of cross-sectional expected returns in terms of economic value, including changes in the importance of the determinants over time.

Due to its flexibility, the $SPPC_p$ can be straightforwardly extended to address transaction costs. For example, we can compute a performance metric based on the gross portfolio return and the return net of transaction costs; the difference between the two metrics is the decrease in performance due to transaction costs. Then, we can use the $SPPC_p$ to decompose the performance metric based on the net portfolio return as well as the change in the performance metric due to transaction costs. The latter measures the contributions of the predictors to the decline in performance due to transaction costs.

Again due to its flexibility, the $SPPC_p$ can also be modified to estimate the predictor contributions to a performance metric when machine learning techniques are used to directly estimate optimal portfolio weights, including those for mean-variance efficient portfolios (e.g., Kozak, Nagel, and Santosh [2020](#); Cong et al. [2022](#); Chen, Pelger, and Zhu [2024](#); Jensen et al. [2026](#)). Since mean-variance efficient portfolio weights relate to the stochastic discount factor, the $SPPC_p$ can provide further insight into the key drivers of the stochastic discount factor.

The voluminous literature on aggregate stock market return predictability considers a lengthy list of predictors, and it would be informative in future research to use the $SPPC_p$ to estimate the contributions of predictors in machine learning models to the economic value generated by aggregate market return predictability. In addition to equities, the $SPPC_p$ can be used to investigate predictor contributions to economic value for portfolios formed from any asset class or combinations of asset classes. In this vein, it would be illuminating to analyze predictor contributions in machine learning models to portfolio performance for a range of asset classes and to explore whether common patterns of predictor importance exist.

Finally, the $SPPC_p$ can potentially be used to improve portfolio performance in real time. For example, at the time of forecast formation, we compute the $SPPC_p$ for each predictor (or group of predictors) for a holdout out-of-sample period. Then, we exclude predictors that contribute negatively to portfolio performance over the holdout out-of-sample period when making return predictions and corresponding asset allocations for the next period. In this way, the $SPPC_p$ serves as a monitoring device as well as a model interpretation tool. We are currently exploring strategies for harnessing the $SPPC_p$ to improve portfolio performance in real time.

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